

(12) **United States Patent**
Cohen et al.

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(54) **FOLDED MACRO-TELE CAMERA LENS DESIGNS INCLUDING SIX LENSES OF +-+--+ OR +--+--, SEVEN LENSES OF ++++--+, OR EIGHT LENSES OF ++++--+ REFRACTIVE POWERS**

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(51) **Int. Cl.**
G02B 13/00 (2006.01)

(52) **U.S. Cl.**
CPC **G02B 13/0065** (2013.01); **G02B 13/0045** (2013.01)

(58) **Field of Classification Search**
CPC G02B 15/142–1425
See application file for complete search history.

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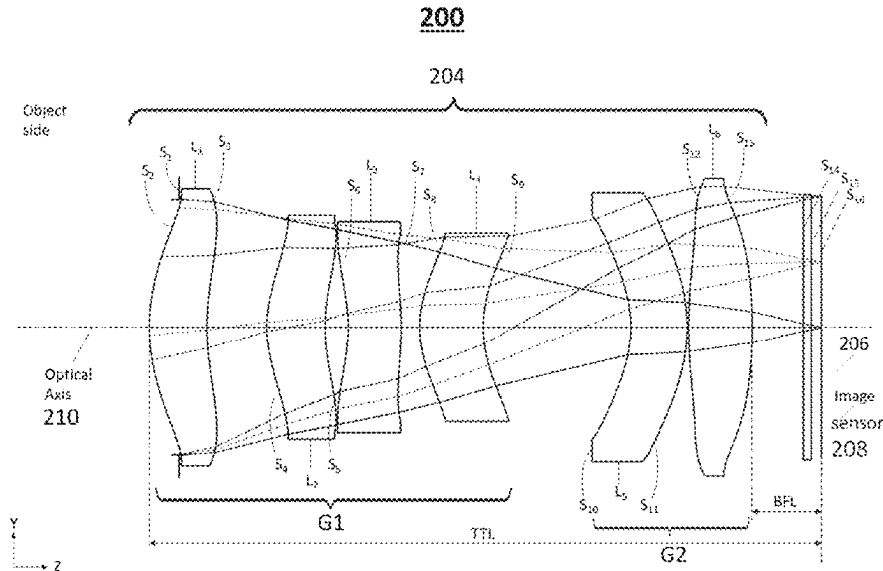
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(57) **ABSTRACT**

Folded digital cameras comprising a lens system with a lens and an image sensor, the lens having $N \geq 6$ lens elements L_i , an effective focal length (EFL) and a total track length (TTL), wherein each lens element has a respective focal length f_i and wherein a first lens element L_1 faces an object side, and an optical path folding element (OPFE) for providing a folded optical path between an object and the lens. In some embodiments, the lens system has a focusing range that covers object-lens distances from infinity to a minimal object distance (MIOD), wherein $MIOD/EFL$ is smaller than 20 or even 7. In some embodiments, the ratio of a maximal chief ray angle to a field of view of the folded camera Max CRA/FOV is smaller than 0.25 or even 0.15 when the camera is focused at infinity.

19 Claims, 12 Drawing Sheets



Related U.S. Application Data

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- (60) Provisional application No. 63/070,501, filed on Aug. 26, 2020, provisional application No. 63/059,200, filed on Jul. 31, 2020.

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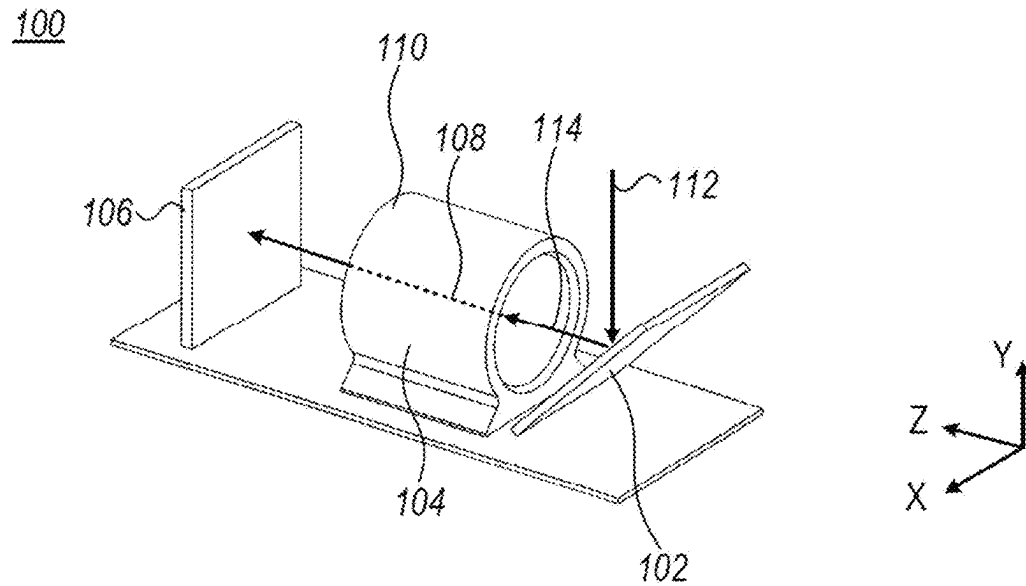


FIG. 1A KNOWN ART

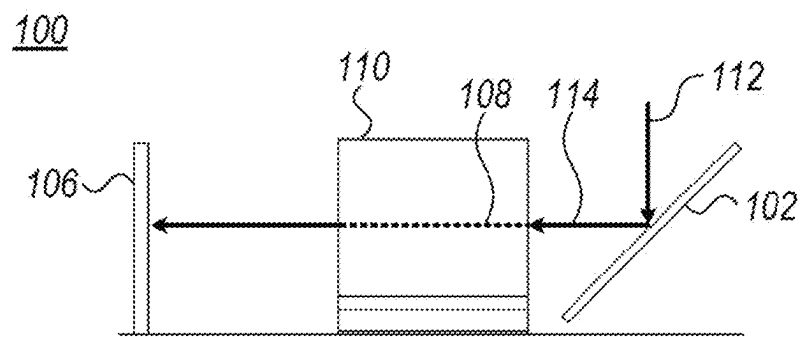
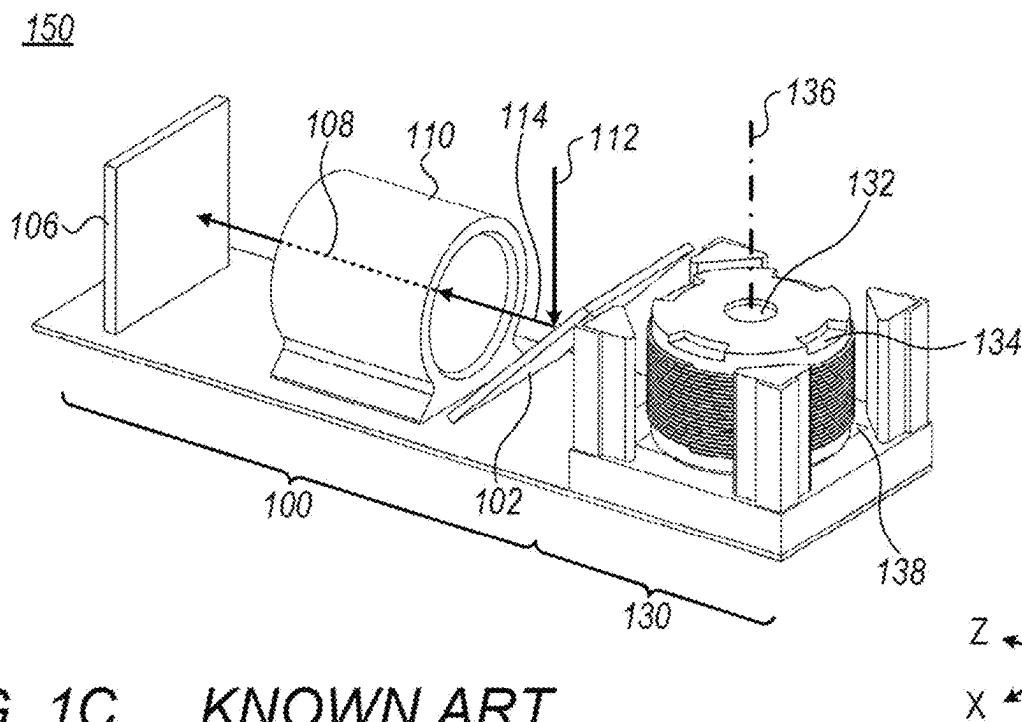


FIG. 1B KNOWN ART



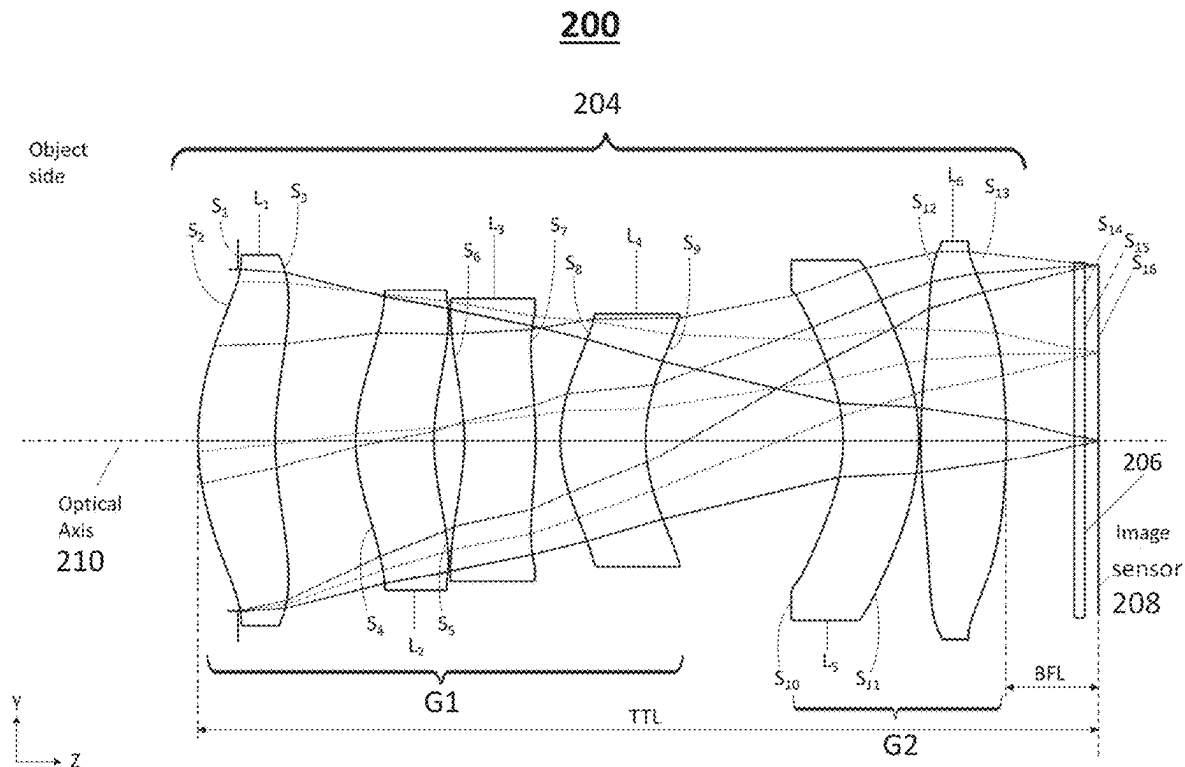


FIG. 2A

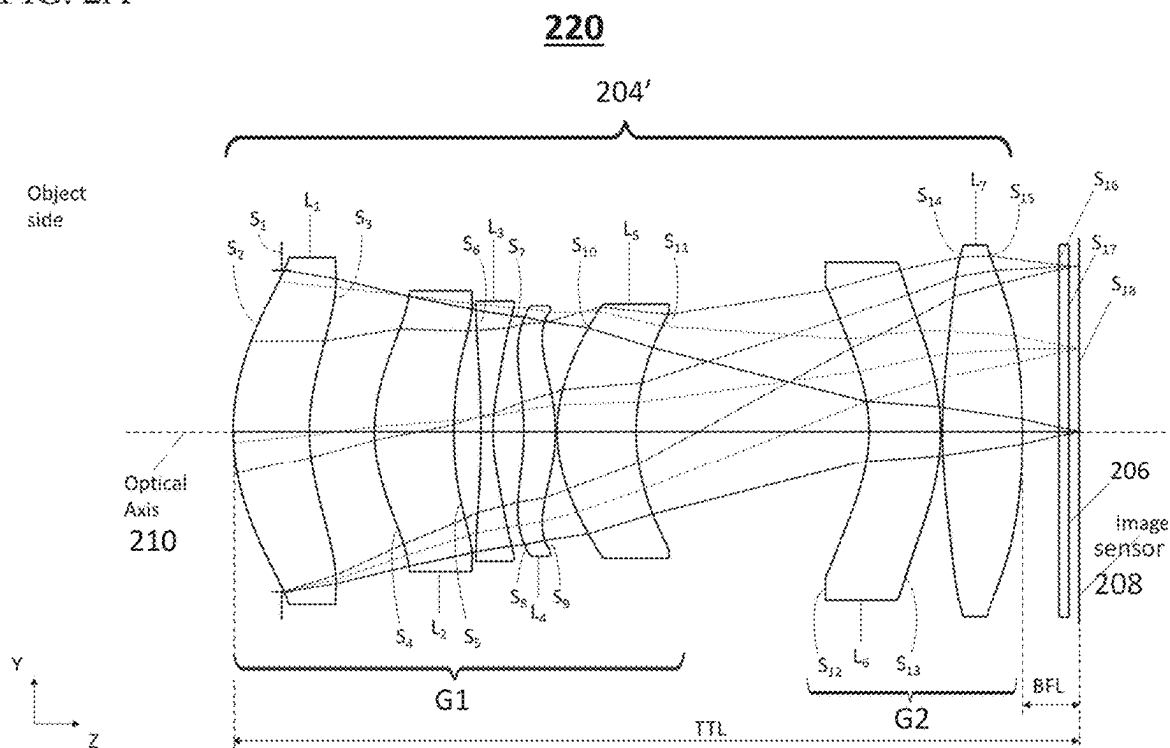


FIG. 2B

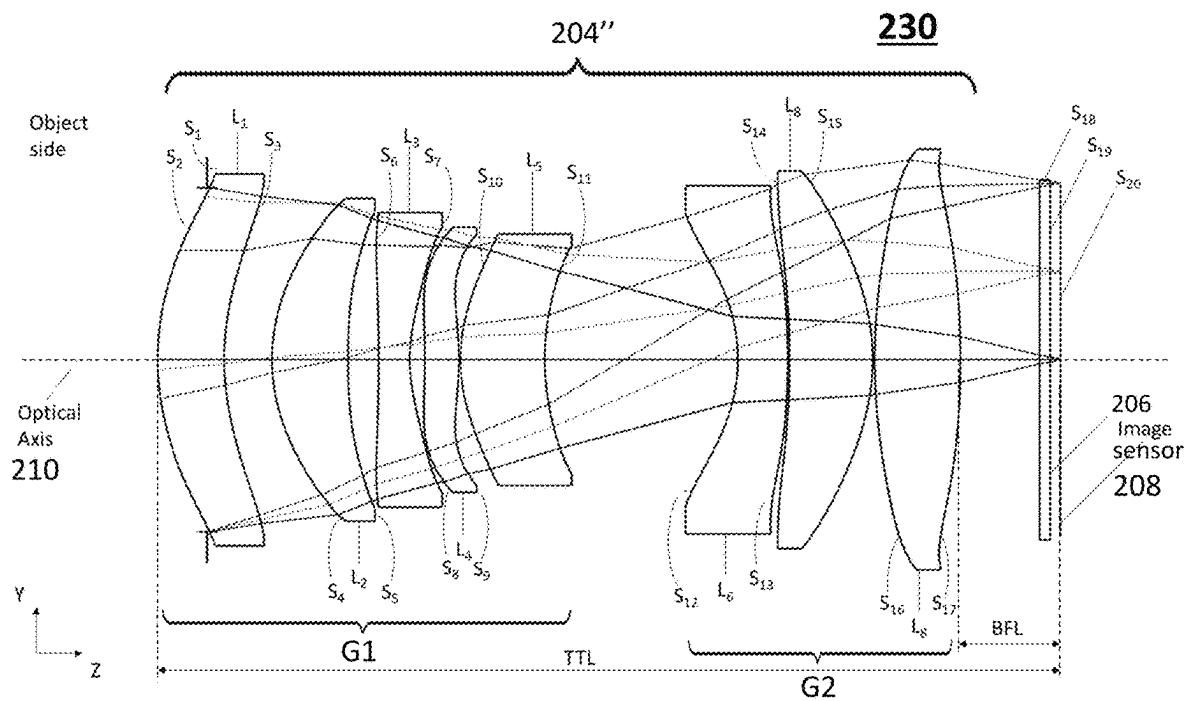


FIG. 2C

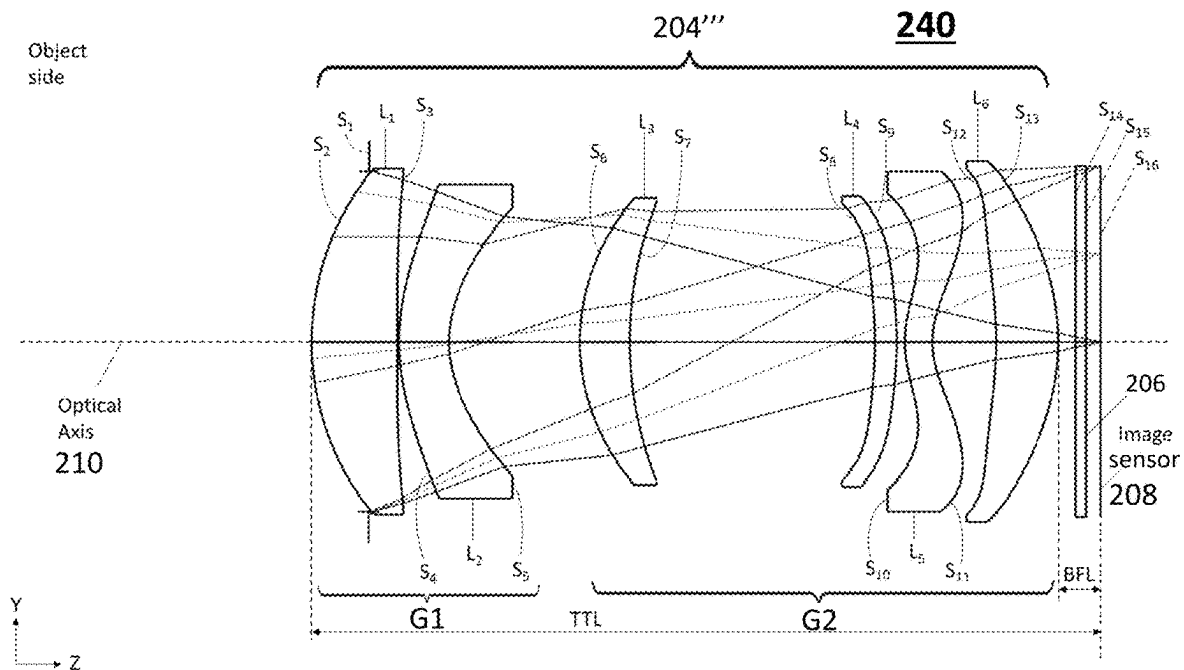


FIG. 2D

240

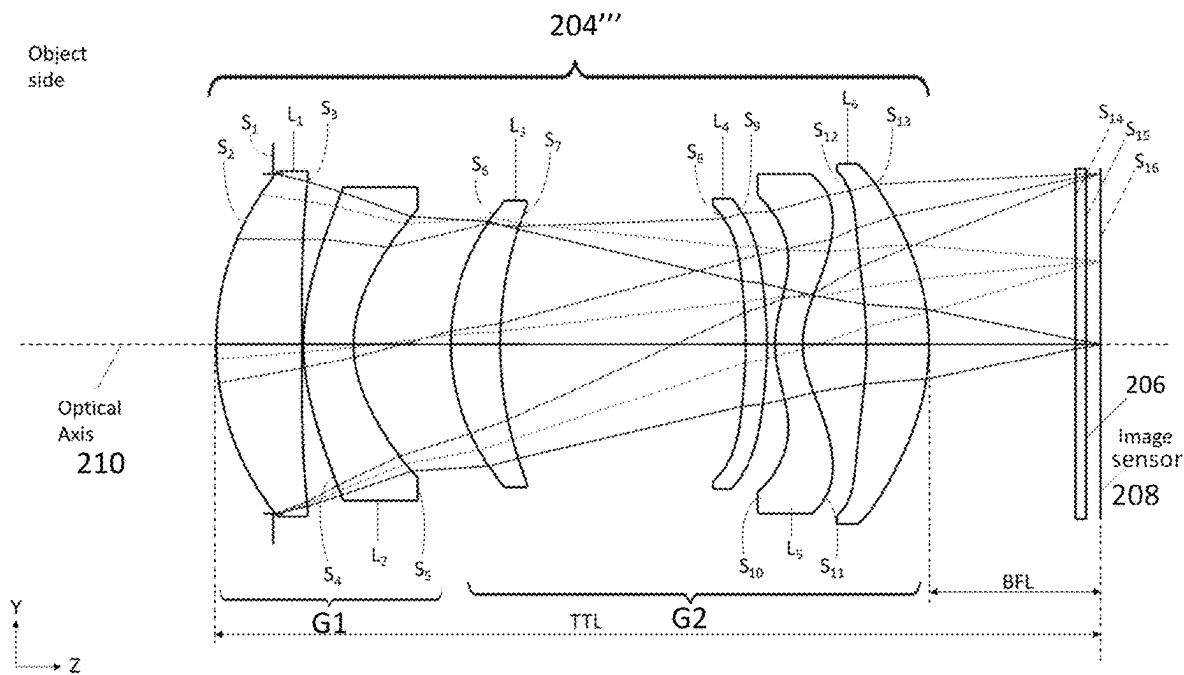


FIG. 2E

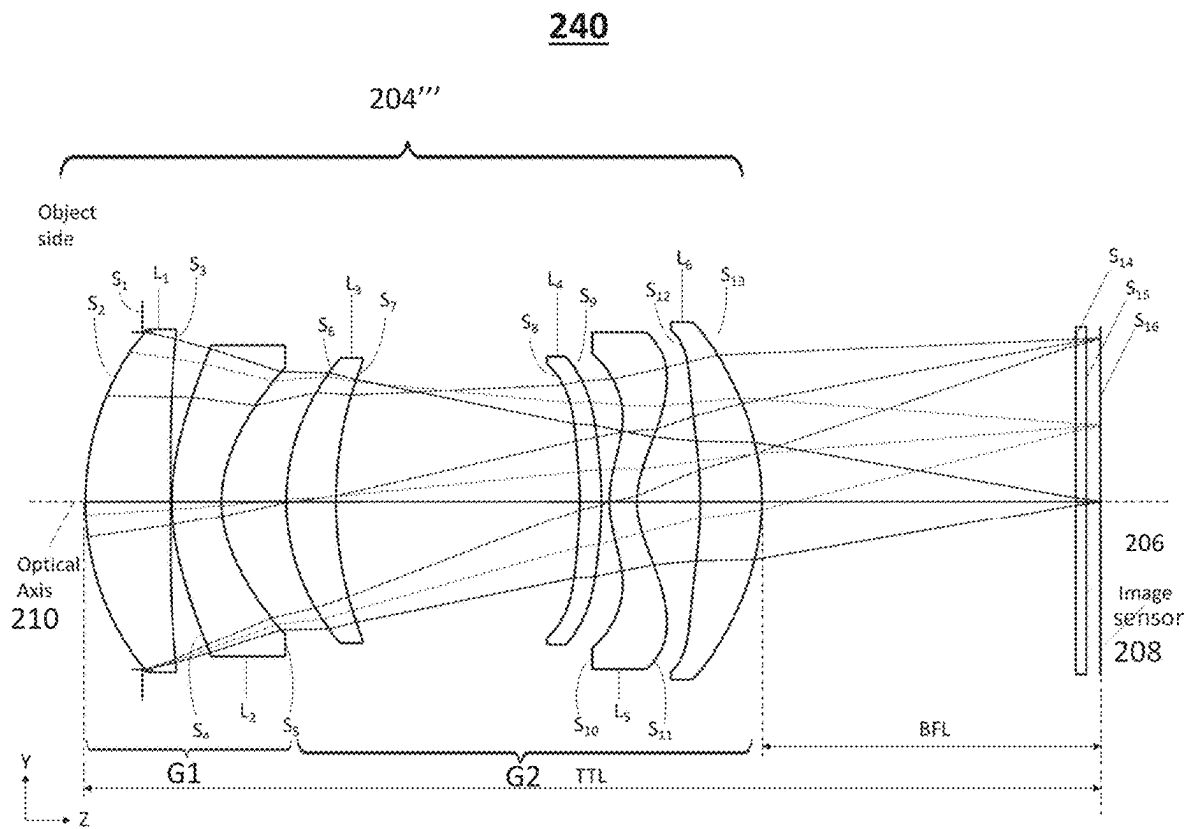


FIG. 2F

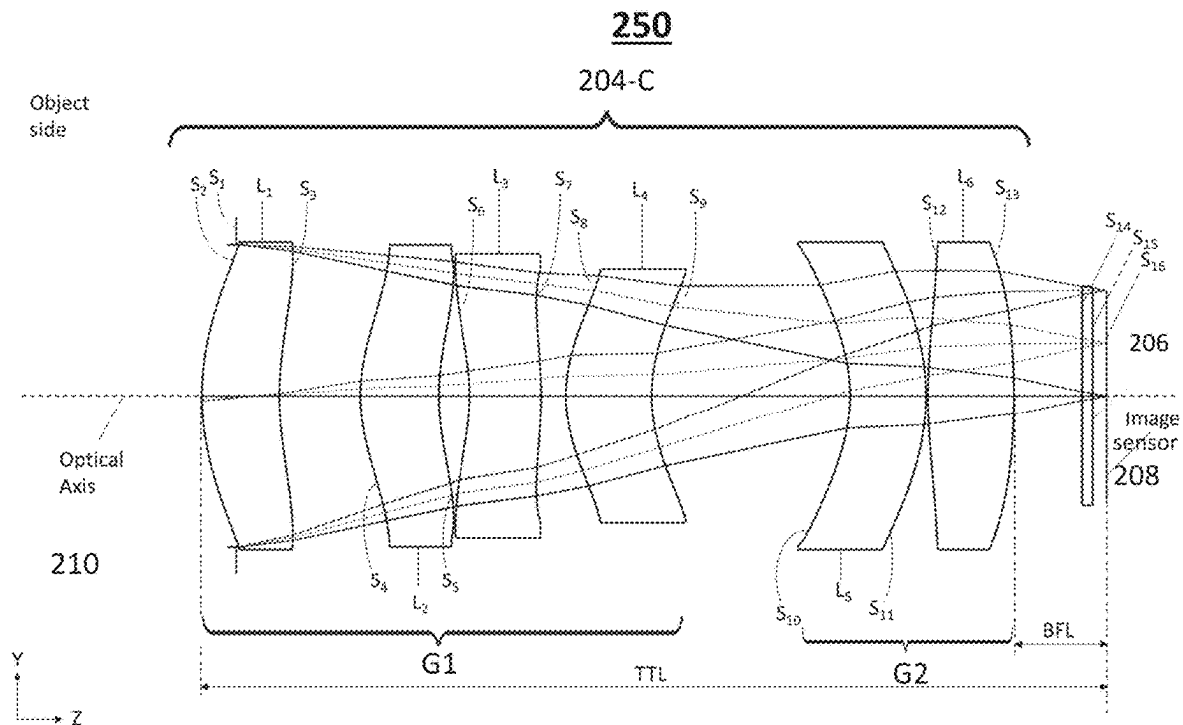


FIG. 2G

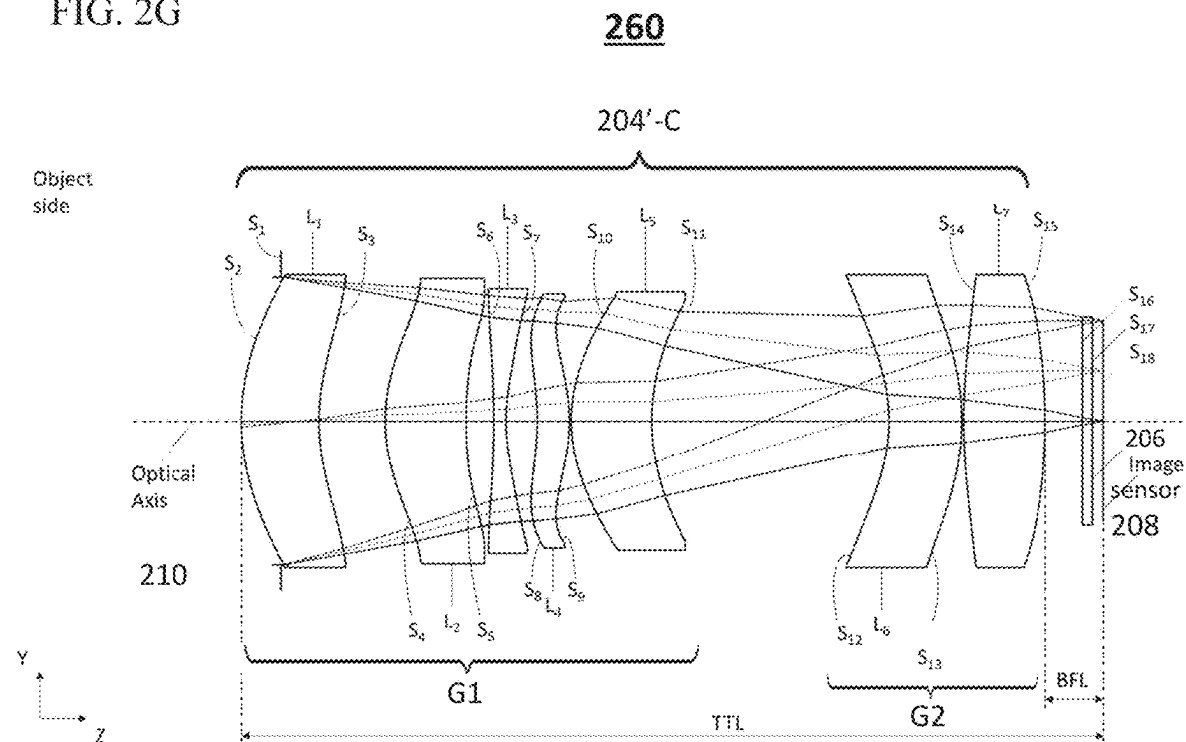


FIG. 2H

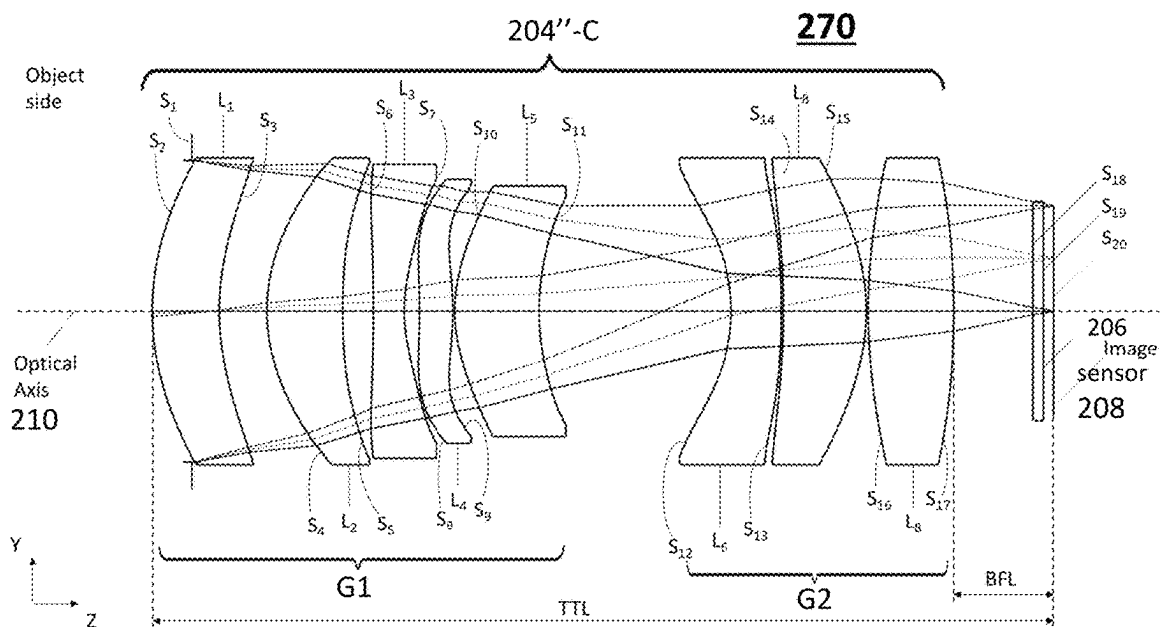


FIG. 2I

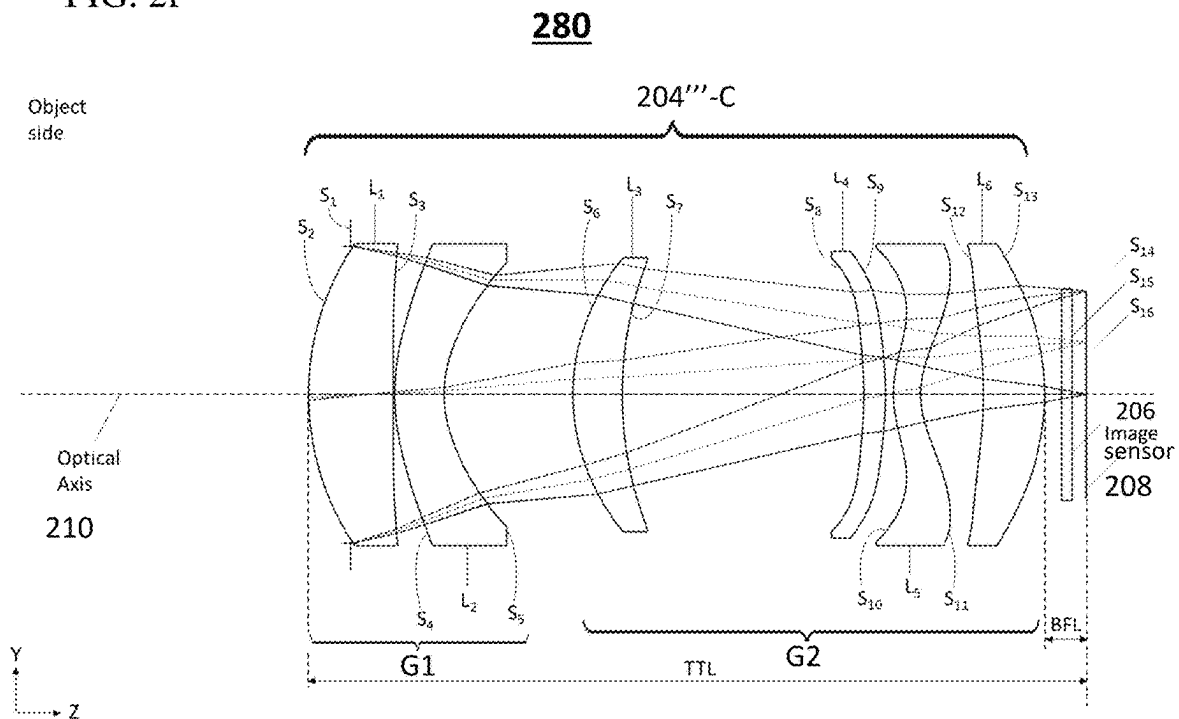


FIG. 2J

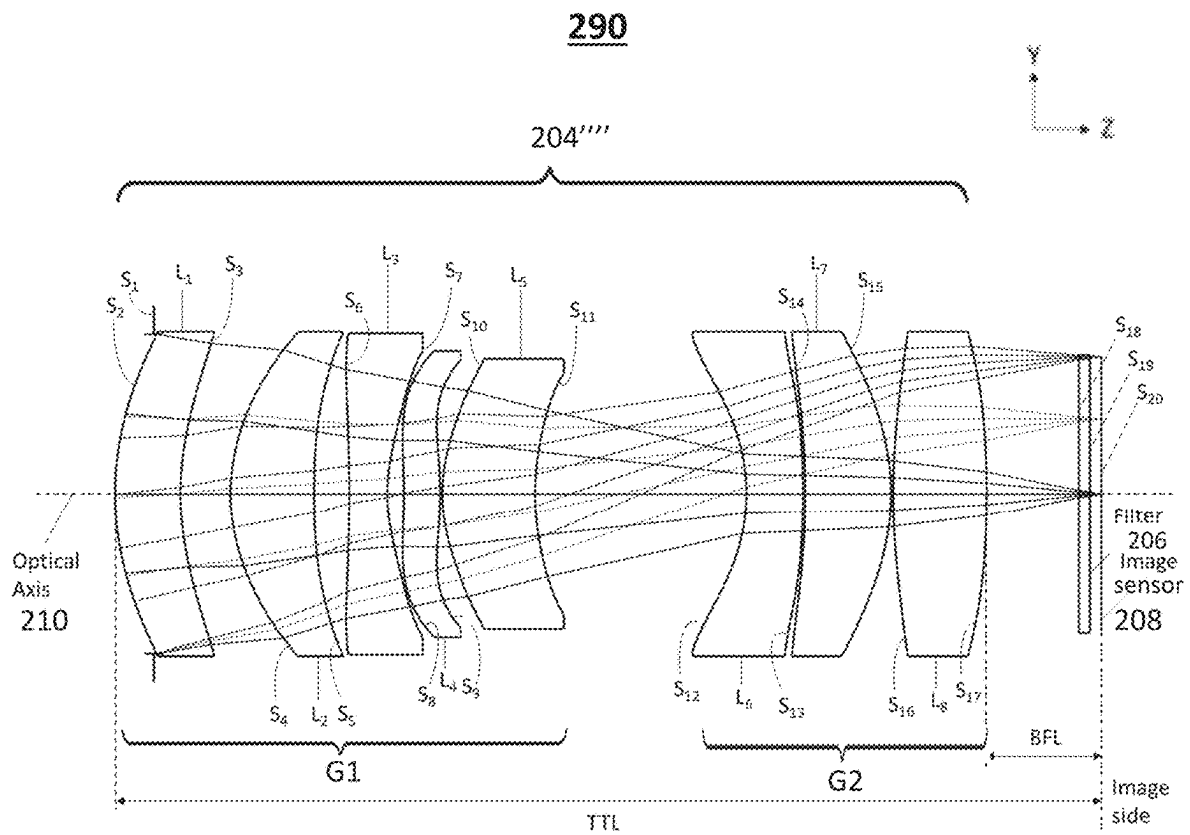


FIG. 2K

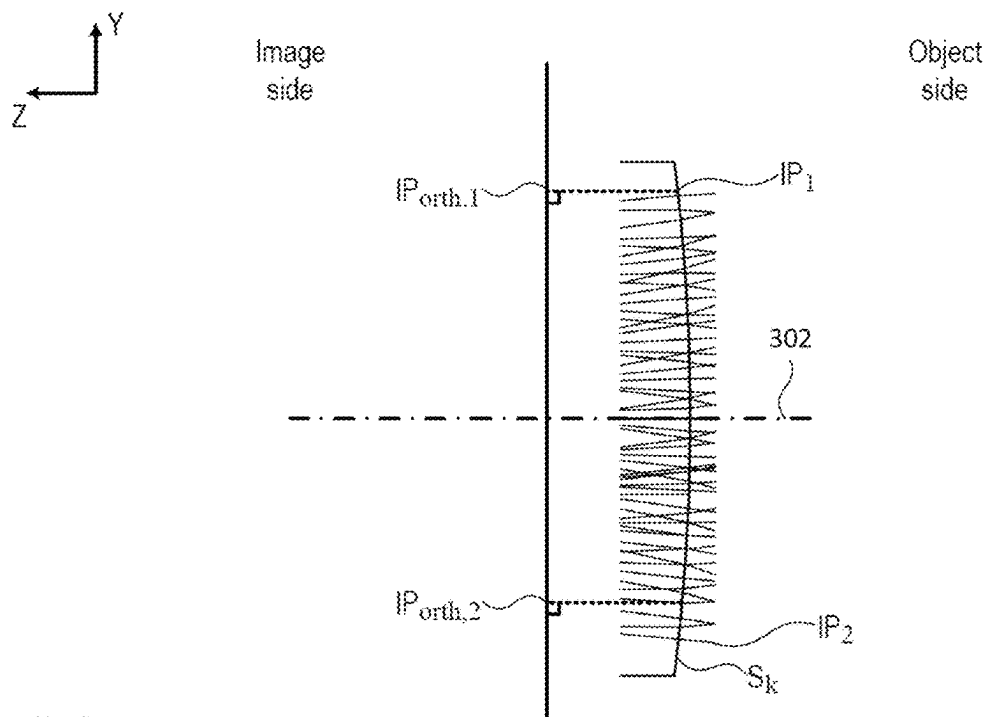


FIG. 3A

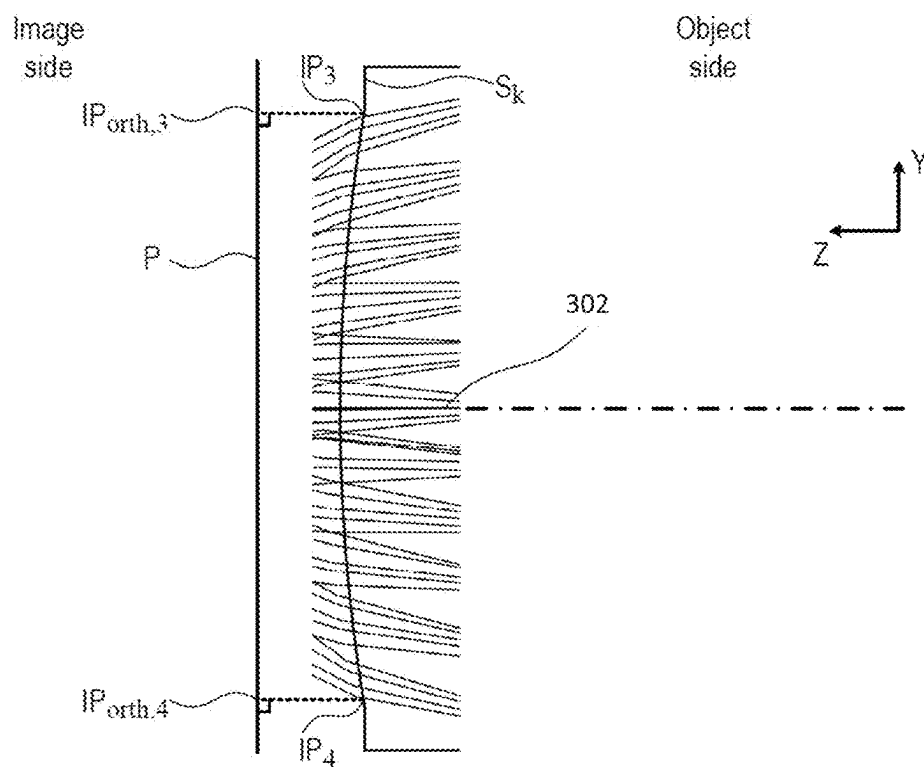


FIG. 3B

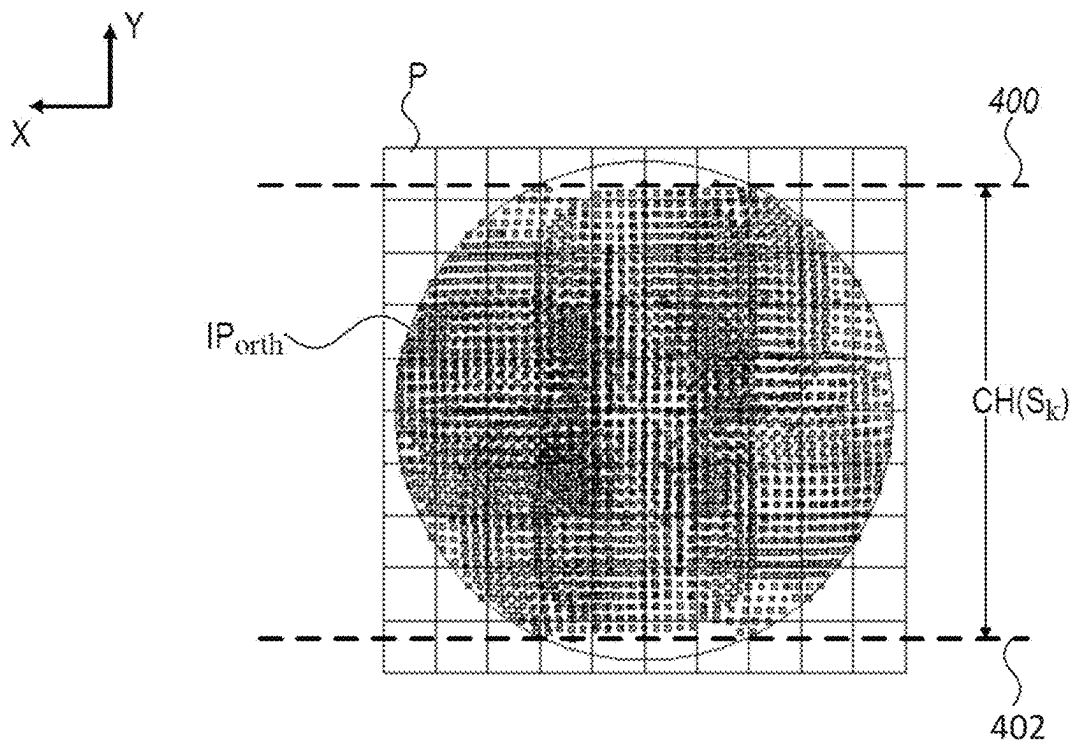


FIG. 4

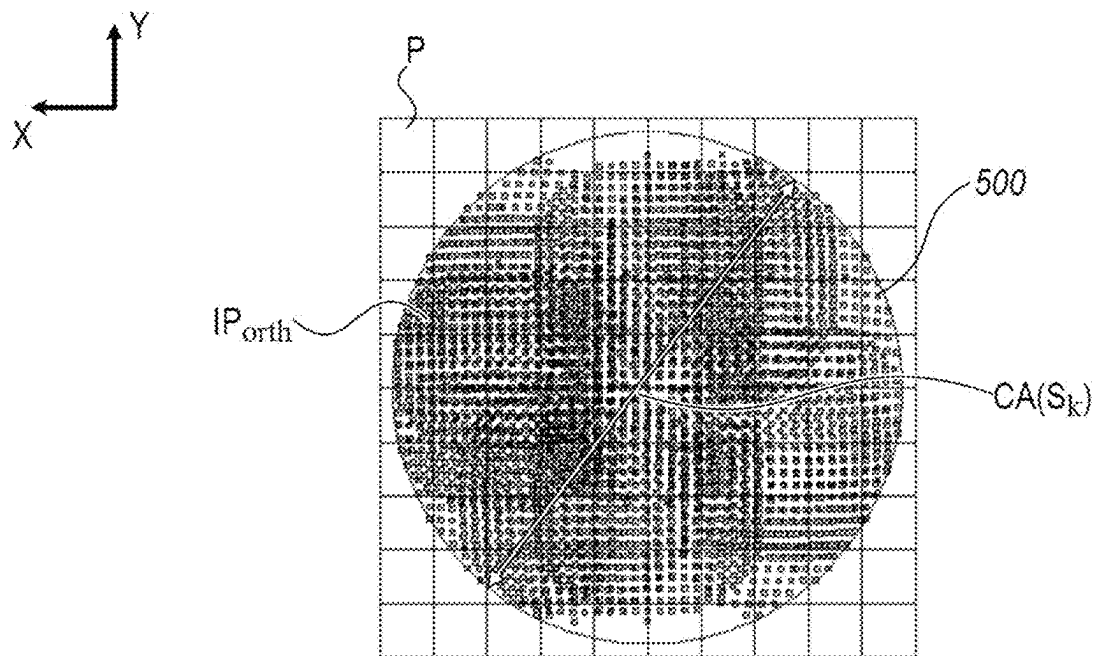


FIG. 5

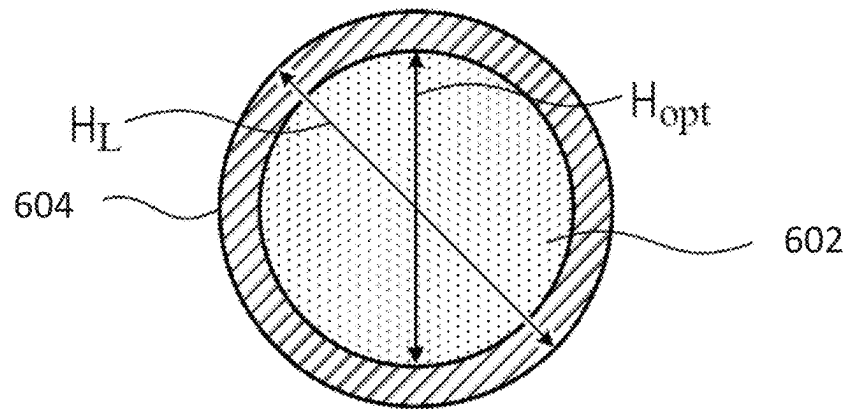


FIG. 6

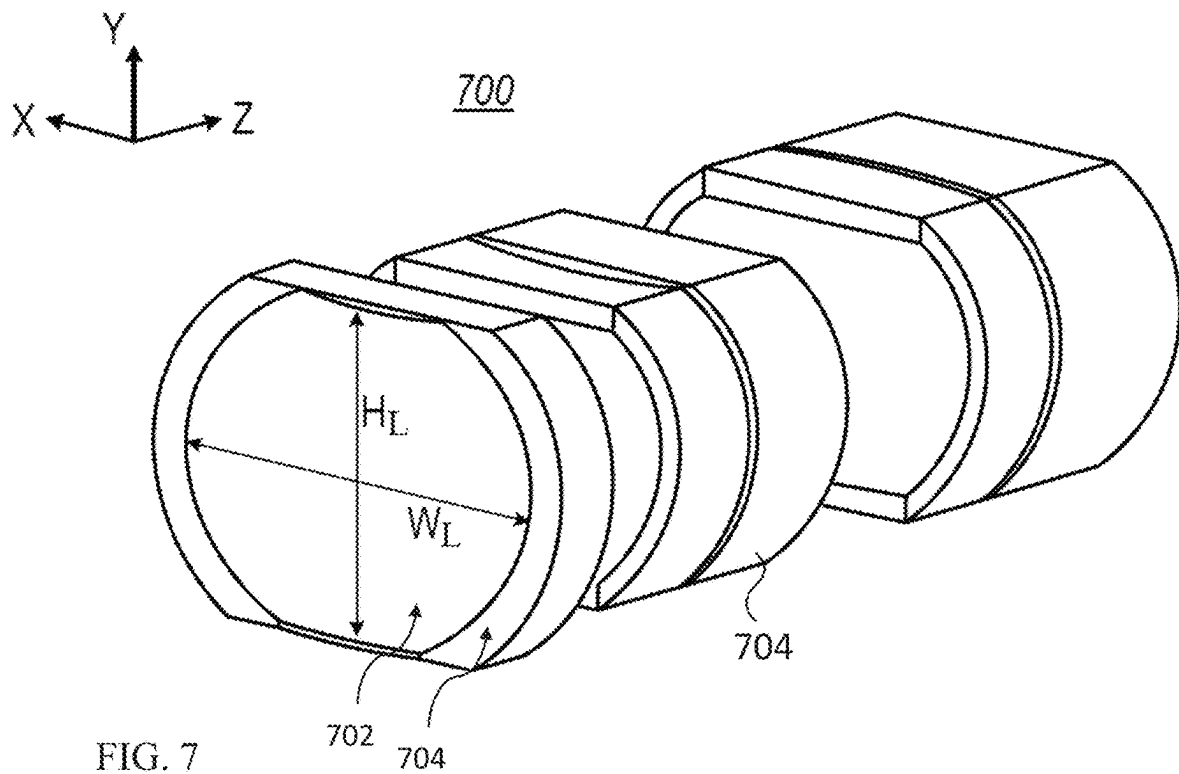


FIG. 7

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**FOLDED MACRO-TELE CAMERA LENS
DESIGNS INCLUDING SIX LENSES OF
+++++ OR +++++, SEVEN LENSES OF
+++++, OR EIGHT LENSES OF ++++++
REFRACTIVE POWERS**

**CROSS REFERENCE TO EXISTING
APPLICATIONS**

This is a continuation from U.S. patent application Ser. No. 18/412,545 filed Jan. 14, 2024 (now allowed), which was a continuation from U.S. patent application Ser. No. 17/614,380 filed Nov. 26, 2021 (now U.S. Pat. No. 11,914,117), which was a 371 application from international patent application PCT/IB2021/056358 filed Jul. 14, 2021, and is related to and claims priority from U.S. Provisional Patent Applications No. 63/059,200 filed Jul. 31, 2020, and 63/070,501 filed Aug. 26, 2020, which are expressly incorporated herein by reference in their entirety.

FIELD

The presently disclosed subject matter is related in general to the field of digital cameras and in particular to folded lenses in such cameras.

DEFINITIONS

In this application and for optical and other properties mentioned throughout the description and figures, the following symbols and abbreviations are used, all for terms known in the art:

Total track length (TTL): the maximal distance, measured along an axis parallel to the optical axis of a lens, between a point of the front surface S_1 of a first lens element L_1 and an image sensor, when the system is focused to an infinity object distance.

Back focal length (BFL): the minimal distance, measured along an axis parallel to the optical axis of a lens, between a point of the rear surface S_{2N} of a last lens element L_N and an image sensor, when the system is focused to an infinity object distance.

Effective focal length (EFL): in a lens (assembly of lens elements L_1 to L_N), the distance between a rear principal point P' and a rear focal point F of the lens.

f-number, (f/#): the ratio of the EFL to an entrance pupil diameter.

Chief ray angle (CRA): a (image-side) chief ray is the line between the center of the exit pupil of a lens system and an image point. The CRA is the angle between a chief ray and the optical axis of the lens. Here and in the following, we refer to image-side chief rays and CRAs, in contrast to an object-side chief ray, which is the line between an object point and the center of the entrance pupil of a lens system. A maximal CRA (Max CRA) is the largest CRA present in a lens system considering all fields supported by the lens system.

BACKGROUND

Multi-aperture cameras (or “multi-cameras”, of which a “dual-cameras” having two cameras is an example) are becoming the standard choice of portable electronic mobile device (e.g. smartphones, tablets, etc.) makers. A multi-camera setup usually comprises a wide field-of-view (or “angle”) FOV_w camera (“Wide” camera or “W” camera), and at least one additional camera, either with the same FOV

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(e.g. a depth auxiliary camera), with a narrower (than FOV_w) FOV (Telephoto or “Tele” camera with FOV_T), or with an ultra-wide field of view FOV_{UW} (wider than FOV_w , “UW” camera).

FIGS. 1A and 1B illustrate a known digital folded camera **100**. Camera **100** comprises an optical path folding element (OPFE) **102** e.g. a prism or mirror, a lens **104** with a plurality of lens elements (not visible in this representation, but visible e.g. in FIGS. 2A-D) and an image sensor **106**. The lens elements may be (as in FIGS. 2A-2D) axial symmetric along an optical axis **108**. In other embodiments, the lens elements may not be axial symmetric. For example, lens elements may be cut (diced, sliced) into a non-circular shape (not shown).

At least some of the lens elements may be included in a “barrel” **110**. The barrel may have a longitudinal symmetry along optical axis **108**. In FIGS. 1A to 1C, the cross-section of this barrel is circular. This is however not mandatory and other shapes can be used, e.g. for hosting cut lens elements.

The path of the optical rays from an object (not shown) to image sensor **106** defines an optical path (see optical paths **112** and **114**, which represent portions of the optical path). OPFE folds the optical path from a first optical path **112** to a second optical path **114**, the latter being substantially parallel to optical axis **108**.

In particular, in some examples, OPFE **102** is inclined at substantially 45 degrees with respect to optical axis **108**. In FIG. 1A, OPFE **102** is also inclined at substantially 45 degrees with respect to optical path **112**.

In some known examples, image sensor **106** lies in a X-Y plane substantially perpendicular to optical axis **108**. This is however not limiting, and image sensor **106** can have a different orientation. For example, and as described in international published patent application WO2016/024192, image sensor **106** may lie in the XZ plane. In this case, an additional OPFE can be used to reflect the optical rays towards image sensor **106**.

Two cameras, for example a Wide camera **100** and a regular UW camera **130** may be included in a digital camera **150** (also referred to as dual-camera). A possible configuration is shown in FIG. 1C. UW camera **130** may include an aperture **132** (indicating the object side of the camera) and an optical lens system **134** (or “Wide lens module”) with a symmetry (and optical) axis **136** in the Y direction, as well as a UW image sensor **138**. The UW camera comprises a UW lens system configured to provide a UW image. As already indicated, the UW camera has a field of view FOV_{UW} larger than FOV_w . For example, FOV_{UW} may be 80-130 degrees and FOV_w may be 60-90 deg. Notably, in other examples, a plurality of Wide cameras and/or a plurality of Tele cameras can be incorporated and operative in a single digital camera. The FOV_T of a Tele camera may be for example 20-50 degrees.

A “Macro-photography” mode is becoming a popular differentiator for smartphone cameras. “Macro-photography” refers to photographing objects close to the camera, so that an image recorded on the image sensor is nearly as large as the actual object photographed. The ratio of image size to object size is the object-to-image magnification M , defined by:

$$M = -\frac{v}{u} = -\frac{\text{dist}(2\text{nd principal plane, Image})}{\text{dist}(\text{Object, 1st principal plane})}$$

where v is a lens-image distance defined by the distance of the 2nd (or “rear”) principal plane of the lens and the image, and u is an object-lens distance defined by the distance of the object to the 1st (or “front”) principal plane of the lens. The minus sign is generally not mentioned explicitly.

In the context of digital single-lens reflex (DSLR) cameras, a Macro image is defined by having a M of about 1:1 or larger, e.g. 1:1.1. In the context of smartphones, “Macro image” may refer to images with M of about 10:1 or even 15:1. First smartphone models have entered the consumer market that provide Macro-photography capabilities, usually by enabling very close focusing with a UW camera, which has a relatively short EFL (e.g. 2.5 mm).

A UW camera can focus to the close range required for Macro photography (e.g., 1.5 cm to 15 cm), but its spatial resolution is poor since its focal length is small and its FOV is large. For example, consider a UW camera with 2.5 mm focal length. When focused to an object at 5 cm (lens-object distance), the UW camera will have approximately $M=19:1$. This according to thin lens equation

$$\frac{1}{EFL} = \frac{1}{u} + \frac{1}{v}$$

with $EFL=2.5$ mm, $v=2.6$ mm and $u=50$ mm. Even when focused to as close as 1.5 cm, the M of the UW camera will be approximately 5:1. Capturing objects in Macro images from these short object-lens distances of e.g. $u=5$ cm or less is very challenging for a user. For example, it may render framing of the image very difficult, it may prohibit taking images of popular Macro objects such as living subjects (e.g. insects), and it may introduce shadows and obscure the lighting in the scene. Additionally, an UW camera has a relatively large depth of field (DoF), even for Macro images. The relatively large DoF corresponds to a low degree of optical Bokeh, which is a highly popular effect in Macro photography.

It would be beneficial to have a Macro camera in mobile devices that captures Macro images from a larger lens-object distance (e.g. 5-15 cm) with larger object to image magnification (e.g. 1:1-15:1), and which has a high degree of optical Bokeh.

SUMMARY

In various embodiments, there are provided folded digital cameras, comprising: a lens system with a lens with $N \geq 6$ lens elements L_i and having an EFL and a TTL, wherein each lens element has a respective focal length f_i and wherein a first lens element L_1 faces an object side; an image sensor; and an OPFE for providing a folded optical path between an object and the lens, wherein the lens system has a focusing range that covers object-lens distances from infinity to a minimal object distance (MIOD), and wherein $MIOD/EFL < 20$.

In various embodiments, there are provided folded digital cameras, comprising: a lens system with a lens with $N \geq 6$ lens elements L_i , having an EFL and a TTL, wherein each lens element L_i has a respective focal length f_i and wherein a first lens element L_1 faces an object side; an image sensor; and an OPFE for providing a folded optical path between an object and the lens, and wherein $Max\ CRA/FOV < 0.25$ when the camera is focused at infinity.

In some embodiments, $MIOD/EFL$ may be smaller than 15, 13, 10, or even than 7.

In some embodiments, the maximum field curvature MFC for any object within the focus range may be smaller than 50 μm .

In some embodiments, a camera may have a f number smaller than 4, 3, or even smaller than 2.5

In some embodiments, the lens elements may be divided into two lens element groups separated by a big gap greater than $TTL/8$, $TTL/7$, or even greater than $TTL/6$.

In some embodiments, the lens elements may be divided into a first lens element group with an effective focal length EFL1 and a second lens element group with an effective focal length EFL2, wherein a ratio $EFL1/EFL2$ may not deviate from 1 by more than 20%, or even by more than 10%

In some embodiments, a camera may further comprise a focusing mechanism for focusing the camera based on a voice coil motor.

In some embodiments, the first lens element L_1 and a second lens element L_2 may be made of a material with an Abbe number greater than 50.

In some embodiments, TTL/EFL may be smaller than 1.5, 1.4, or even smaller than 1.3.

In some embodiments, a ratio of the focal length f_N of the last lens element L_N and the TTL, f_N/TTL may be smaller than 1.0, 0.9, 0.8, 0.75, or even smaller than 0.7.

In some embodiments, $Max\ CRA/FOV$ may be smaller than 0.2, or even smaller than 0.15.

In some embodiments, a maximum field curvature MFC for any object within the focus range may be smaller than 50 μm .

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting examples of embodiments disclosed herein are described below with reference to figures attached hereto that are listed following this paragraph. The drawings and descriptions are meant to illuminate and clarify embodiments disclosed herein, and should not be considered limiting in any way.

FIG. 1A illustrates schematically a known digital folded camera in a perspective view;

FIG. 1B illustrates schematically the camera of FIG. 1A in a side view;

FIG. 1C shows schematically a known dual-camera including a digital camera and an upright camera in a perspective view;

FIG. 2A shows schematically an optical lens system disclosed herein;

FIG. 2B shows schematically another optical lens system disclosed herein;

FIG. 2C shows schematically yet another optical lens system disclosed herein;

FIG. 2D shows schematically yet another optical lens system disclosed herein, focused to infinity;

FIG. 2E shows the lens system of FIG. 2D focused to 100 mm;

FIG. 2F shows the lens system of FIG. 2D focused to 50 mm;

FIG. 2G shows schematically an optical lens system disclosed herein with lens elements cut to a non-circular shape;

FIG. 2H shows schematically another optical lens system disclosed herein with lens elements cut to a non-circular shape;

FIG. 2I shows schematically yet another optical lens system disclosed herein with lens elements cut to a non-circular shape;

FIG. 2J shows schematically yet another optical lens system disclosed herein with lens elements cut to a non-circular shape;

FIG. 2K shows schematically yet another optical lens system disclosed herein, focused to infinity;

FIG. 3A illustrates the orthogonal projections $IP_{orth,1}$, $IP_{orth,2}$ of two impact points IP_1 and IP_2 on a plane P that is orthogonal to the optical axis of the lens of the system in FIGS. 2A-2D;

FIG. 3B illustrates the orthogonal projections $IP_{orth,3}$, $IP_{orth,4}$ of two impact points IP_3 and IP_4 on a plane P that is orthogonal to the optical axis of the lens of the system in FIGS. 2A-2D;

FIG. 4 provides graphically a definition of clear height;

FIG. 5 provides graphically a definition of clear aperture;

FIG. 6 provides a graphical illustration of diameter H_{Li} of lens element L_i ;

FIG. 7 shows an exploded view of lens elements illustrating lens element width W_L and height H_L .

DETAILED DESCRIPTION

In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding. However, it will be understood by those skilled in the art that the presently disclosed subject matter may be practiced without these specific details. In other instances, well-known methods have not been described in detail so as not to obscure the presently disclosed subject matter.

FIGS. 2A-2J illustrate various lens systems disclosed herein. All lens systems shown in FIG. 2A-J can be included in a folded camera such as shown in FIGS. 1A and 1B.

FIG. 2A shows schematically an optical lens system disclosed herein and numbered 200. Lens system 200 comprises a lens 204, an optical element 206 and an image sensor 208. System 200 is shown with ray tracing. Optical element 206 may be for example an infra-red (IR) filter, and/or a glass image sensor dust cover. Optical rays (after their reflection by prism 202) pass through lens 204 and form an image on image sensor 208. FIG. 2A shows 3 fields with 3 rays for each: the upper marginal-ray, the lower marginal-ray and the chief-ray. In the example of FIG. 2A, the optical rays pass through optical element 206 before impinging on image sensor 208. This is however not limiting, and in some examples, optical element 206 is not present. That is, the optical element may be optional in some designs.

Lens 204 includes a plurality of N lens elements L_i where “i” is an integer between 1 and N. In lens system 200, N is equal to six. This is however not limiting and a different number of lens elements can be used. According to some examples, N is equal to or greater than 5. For example, N can be equal to 5, 6, 7, 8, 9 or 10. L_1 is the lens element closest to the object (prism) side and L_N is the lens element closest to the image side, i.e. the side where the image sensor is located. This order holds for all lenses and lens elements disclosed herein. Lens elements L_i can be used e.g. as lens elements of a camera similar to camera 100. The N lens elements are axial symmetric along an optical (lens) axis 210. Each lens element L_i comprises a respective front surface S_{2i-1} (the index “2i-1” being the number of the front surface) and a respective rear surface S_{2i} (the index “2i” being the number of the rear surface). This numbering convention is used throughout the description. Alternatively, as done throughout this description, lens surfaces are marked as “ S_k ”, with k running from 1 to 2N. The front surface and the rear surface can be in some cases aspherical. This is however not limiting.

As used herein, the term “front surface” of each lens element refers to the surface of a lens element located closer to the entrance of the camera (camera object side), and the

term “rear surface” refers to the surface of a lens element located closer to the image sensor (camera image side).

As explained below, a clear height value $CH(S_k)$ and a clear aperture value $CA(S_k)$ can be defined for each surface S_k (for $1 \leq k \leq 2N$). $CA(S_k)$ and $CH(S_k)$ define optical properties of each surface S_k of each lens element. The CH term is defined with reference to FIG. 4 and the CA term is defined with reference to FIG. 5.

In addition a height H_{Li} (for $1 \leq i \leq N$) is defined for each lens element L_i . H_{Li} corresponds, for each lens element L_i , to the maximal height of lens element L_i measured along an axis perpendicular to the optical axis of the lens elements. For a given lens element, the respective H_{Li} is greater than, or equal to the CH and the CA of the front and rear surfaces of this given lens element. Typically, for an axial symmetric lens element, H_{Li} is the diameter of lens element L_i as seen in FIG. 6. Typically, for an axial symmetric lens element, $H_{Li} = \max\{CA(S_{2i-1}), CA(S_{2i})\} + \text{a mechanical part size}$. In general, in lens design the mechanical part size is defined as not contributing to the optical properties of the lens. Consequently, one defines two heights of a lens element L_i , an optical height $H_{opt,i}$ (corresponding to the CA value) of an optically active area 602 and a geometrical height of the lens H_{Li} of an entire lens area 604 which covers an optically active and an optically inactive area. The mechanical part and its properties are defined below. The mechanical part size contribution to H_{Li} is typically 200-1000 μm .

In lens system 200, some of the surfaces of the lens elements are represented as convex, and some are represented as concave. The representation of FIG. 2A is however not limiting, and a different combination of convex and/or concave surfaces can be used, depending on various factors such as the application, the desired optical power, etc.

As shown in FIGS. 3A, 3B and 4, each optical ray that passes through a surface S_k impinges this surface on an impact point IP. Optical rays enter optical lens system 200 from surface S_1 and pass through surfaces S_2 to S_{2N} . Some optical rays can impinge on any surface S_k but cannot/will not reach image sensor 208. For a given surface S_k , only optical rays that can form an image on image sensor 208 are considered. $CH(S_k)$ is defined as the distance between two closest possible parallel lines, see lines 400 and 402 in FIG. 4 located on a plane P orthogonal to the optical axis of the lens elements. In the representation of FIGS. 3A and 3B, plane P is parallel to plane X-Y and is orthogonal to optical axis 302 such that the orthogonal projection IP_{orth} of all impact points IP on plane P is located between the two parallel lines.

The definition of $CH(S_k)$ does not depend on the object currently imaged, since it refers to the optical rays that “can” form an image on the image sensor. Thus, even if the currently imaged object is located in a black background that does not produce light, the definition does not refer to this black background since it refers to any optical rays that “can” reach the image sensor to form an image (for example optical rays emitted by a background which would emit light, contrary to a black background).

For example, FIG. 3A illustrates the orthogonal projections $IP_{orth,1}$, $IP_{orth,2}$ of two impact points IP_1 and IP_2 on plane P which is orthogonal to optical axis 302. By way of example, in the representation of FIG. 3A, surface S_k is convex.

FIG. 3B illustrates the orthogonal projections $IP_{orth,3}$, $IP_{orth,4}$ of two impact points IP_3 and IP_4 on plane P. By way of example, in the representation of FIG. 3B, surface S_k is concave.

In FIG. 4, the orthogonal projection IP_{orth} of all impact points IP of a surface S_k on plane P is located between parallel lines 400 and 402. $CH(S_k)$ is thus the distance between lines 400 and 402.

Attention is drawn to FIG. 5. According to the presently disclosed subject matter, a clear aperture $CA(S_k)$ is defined for each given surface S_k (for $1 \leq k \leq 2N$), as the diameter of a circle, wherein the circle is the smallest possible circle located in a plane P orthogonal to the optical axis **108** and

for refraction index ("Index") and Abbe #. Each lens element L_i has a respective focal length f_i , given in Table 1. The FOV is given as half FOV (HFOV). The definitions for surface types, Z axis, CA values, reference wavelength, units, focal length and HFOV are valid for Tables 1-17.

TABLE 1

Lens system 200 EFL = 15.0 mm, F number = 2.2, Half FOV = 13.25 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S	Plano	Infinity	-0.795	3.408				
2	Lens 1	ASP	5.103	1.539	3.645	Plastic	1.54	55.91	35.74
3			6.170	1.607	3.606				
4	Lens 2	ASP	3.492	1.559	2.944	Plastic	1.54	55.91	17.63
5			4.611	0.620	2.792				
6	Lens 3	ASP	-4.169	1.421	2.772	Plastic	1.67	19.44	-14.69
7			-8.176	0.493	2.475				
8	Lens 4	ASP	3.021	1.701	2.469	Plastic	1.54	55.91	20.11
9			3.337	3.954	2.438				
10	Lens 5	ASP	-3.359	1.509	2.942	Plastic	1.67	19.44	-39.43
11			-4.540	0.040	3.543				
12	Lens 6	ASP	12.089	1.704	3.920	Plastic	1.54	55.91	12.05
13			-13.762	1.368	3.896				
14	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.17	
15			Infinity	0.275	—				
16	Image	Plano	Infinity	—	—				

encircling all orthogonal projections IP_{orth} of all impact points on plane P. As mentioned above with respect to $CH(S_k)$, the definition of $CA(S_k)$ also does not depend on the object which is currently imaged. As shown in FIG. 5, the circumscribed orthogonal projection IP_{orth} of all impact points IP on plane P is a circle **500**. The diameter of circle **500** defines $CA(S_k)$.

Detailed optical data and surface data are given in Tables 1-3 for the example of the lens elements in FIG. 2A. The values provided for these examples are purely illustrative, and according to other examples other values can be used.

Surface types are defined in Table 1. The coefficients for the surfaces are defined in Table 2. The surface types are:

- Plano: flat surfaces, no curvature
- Q type 1 (QT1) surface sag formula:

$$z(r) = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + D_{con}(u) \quad (\text{Eq. 1})$$

$$D_{con}(u) = u^4 \sum_{n=0}^N A_n Q_n^{con}(u^2)$$

$$u = \frac{r}{r_{norm}}, \quad x = u^2$$

$$Q_0^{con}(x) = 1 \quad Q_1^{con} = -(5 - 6x) \quad Q_2^{con} = 15 - 14x(3 - 2x)$$

$$Q_3^{con} = -[35 - 12x[14 - x(21 - 10x)]]$$

$$Q_4^{con} = 70 - 3x[168 - 5x[84 - 11x(8 - 3x)]]$$

$$Q_5^{con} = -[126 - x(1260 - 11x[420 - x[720 - 13x(45 - 14x)]])]$$

where $\{z, r\}$ are the standard cylindrical polar coordinates, c is the paraxial curvature of the surface, k is the conic parameter, r_{norm} is generally one half of the surface's clear aperture, and A_n are the polynomial coefficients shown in lens data tables. The Z axis is positive towards image. Values for CA are given as a clear aperture radius, i.e. $CA/2$. The reference wavelength is 555.0 nm. Units are in mm except

TABLE 2

Aspheric Coefficients					
Surface #	Conic A4	A6	A8	A10	
2	0	-2.64E-03	-8.84E-07	2.34E-07	-5.05E-07
3	0	-9.49E-03	6.22E-04	-3.76E-05	7.07E-07
4	0	-1.37E-02	-2.75E-04	9.19E-05	-7.12E-06
5	0	-1.48E-02	9.78E-04	-8.57E-05	2.50E-06
6	0	2.37E-02	-2.66E-03	2.23E-04	-7.42E-06
7	0	1.36E-02	-1.29E-03	8.14E-05	3.27E-06
8	0	-1.13E-02	-7.97E-04	4.89E-05	-8.25E-06
9	0	-8.28E-03	2.76E-05	-1.17E-04	5.77E-06
10	0	1.01E-02	-2.86E-04	5.62E-05	-2.92E-06
11	0	4.11E-03	-4.26E-04	6.78E-05	-3.01E-06
12	0	-2.04E-03	-2.68E-04	3.25E-05	-7.07E-07
13	0	-1.26E-03	-7.29E-05	-2.48E-06	6.02E-07

Table 3 provides details on the variation of the properties of lens system **200** with the object-lens distance. The object-lens distance is defined by the distance of the object to the 1st principal plane of the lens.

Table 4 provides details on the maximum (image-side) CRAs of lens system **200**. The maximum CRA and Half FOV (HFOV) are given for several object-lens distances ("Object"). Data refers to a field of 3.5 mm, corresponding to an edge of the image sensor (i.e. upper end of sensor diagonal).

For achieving small values of maximum CRA, the focal length f_N of the last lens element L_N is smaller than the lens' TTL. The TTL of lens **204** is 18 mm. For lens **204**, $f_6=12.05$ mm and a ratio of $f_N/\text{TTL}=\mathbf{0.67}$.

TABLE 3

Lens system 200 Variation of lens properties with object distance			
Object Distance [mm]	BFL [mm]	HFOV [deg]	Magnification
Infinity	1.853	13.25	0.000
1000	2.076	13.17	-0.015
500	2.299	13.08	-0.030
300	2.599	12.98	-0.050
250	2.750	12.92	-0.060
200	2.977	12.83	-0.076
150	3.357	12.67	-0.101
100	4.125	12.34	-0.153

TABLE 4

Max CRA/FOV-Lens system 200			
Object [mm]	Max CRA	HFOV (Image Height ("IH") = 3.5 mm) [deg]	Max CRA/FOV (IH = 3.5 mm)
Infinity	4.96	13.29	0.19
1000	5.59	13.22	0.21
500	6.09	13.14	0.23
300	6.02	13.06	0.23
250	5.76	13.01	0.22
200	5.87	12.93	0.23
150	6.02	12.80	0.24
100	6.22	12.51	0.25

The focusing range of lens system **200** is from infinity to 100 mm. The focusing range of a lens system is defined as all object-lens distances that can be focused to by means of a camera mechanism that controls the distance between lens and image sensor. That is, for each object located within the focus range, a focusing mechanism can set a particular lens-image sensor distance that results in maximum contrast for the object's image. Maximum contrast means that for lens-image sensor distances other than the particular lens-image sensor distance, the object's contrast will decrease. The minimal object distance (MIOD) is defined as the lower limit of the focusing range, i.e. the MIOD is the smallest object-lens distance that the lens system can focus to. For lens system **200**, the MIOD is 100 mm. Lens system **200** can focus continuously from infinity to 100 mm, i.e. any focus position between Infinity to 100 mm (as well as any magnification between 0 and -0.153) can be realized.

For focusing lens **204**, all lens elements are moved together. For changing focus from infinity to 100 mm, a lens movement ("lens stroke") of 2.272 mm is required. For moving the lens, an actuator as known in the art may be used, e.g. a voice coil motor (VCM). A Hall sensor-magnet geometry for large stroke linear position sensing which is required for VCMs supporting large strokes such as 2 mm or more is described in the U.S. Provisional Patent Application No. 63,059,200. At the MIOD, lens system **200** achieves a magnification of -0.153, corresponding to an object-image ratio of ca. 6.5:1. The HFOV decreases from 13.25 degrees when focused to infinity to 12.34 degrees when focused to the MIOD.

For any object within the focus range, lens system **200** has a maximum field curvature (MFC) smaller than 50 μm . MFC may be defined as follows: when placing a lens such as lens **204** at a distance v from a flat image sensor such as image sensor **208**, image points at the optical axis will be in perfect

focus, but image points off the optical axis will come into focus not at the image sensor, but at a distance v' smaller than v , wherein v' is less than MFC for all image points.

A lens such as lens **204** can be divided into two lens groups, a first lens group ("focusing group" or "G1") and a second lens group ("CRA correction group" or "G2"). In lens **204**, the focusing group includes lens elements L_1 , L_2 , L_3 and L_4 . The CRA correction group includes L_4 and L_5 . The focusing group and the CRA correction group are separated spatially from each other by a big gap (BG) of 3.954 mm. All lens elements of G1 together have an EFL1=14.71 mm. All lens elements of G2 together have an EFL2=13.55 mm.

In another lens system embodiment **250** shown in FIG. 2G and for achieving a folded lens system with low $f/\#$ and low lens height at the same time, lens elements are cut to a non-circular shape ("D-cut"). The cut lens elements may be obtained by cutting the large lens elements of lens **204** to a height of e.g. 6 mm (in Y direction), so that the lens is now a cut lens **204-C**. That is, the lens elements L_1 of lens **204** that have height $H_{Li} > 6$ mm (i.e. L_1 , L_5 and L_6) are cut to 6 mm. Lens elements are cut in the Y-direction. The cut is performed symmetrically, i.e. the cut top part (defined by larger Y values in FIG. 2A) is identical in size to the cut bottom (defined by smaller Y values in FIG. 2A). The cut lens elements in **204-C** have no circular symmetry like that of lens elements **204**; their width is larger than their height, i.e. $W_{Li} > H_{Li}$ (see example in FIG. 7). Lens elements L_1 of lens **204** that have height $H_{Li} \leq 6$ mm are not changed. As to the cut, cut lens elements L_1 , L_5 and L_6 have a large CA but still low CH. This is beneficial as in a folded lens design in which the camera height is limited in general by a height of the host device. A folded lens having large CA and low CH is beneficial for having a low $f/\#$ folded lens that is compatible with e.g. a smartphone's height constraints. The CA of lens elements L_2 , L_3 and L_4 may be oriented in any direction, e.g. in Y direction. As of the cut design, the CA of lens element L_1 , L_5 and L_6 is oriented in the X direction (not visible here).

Light is lost at extreme rays by cutting the lens elements, but no light loss is expected for center rays. Light loss is given as percentage of rays-through at the image plane at an image sensor boundary having coordinates (X, Y)=(0, 2.1 mm), i.e. moving up from the optical axis by 2.1 mm:

Infinity: 92.49%
1 m: 92.22%
0.5 m: 91.94%
20 cm: 91.01%
10 cm: 89.43%

In other embodiments, only one or only two lens elements L_1 may be cut, i.e. may have $W_{Li} > H_{Li}$. In yet other embodiments, more than three lens elements L_1 may be cut, i.e. may have $W_{Li} > H_{Li}$. In yet other embodiments, all lens elements L_1 may be cut, i.e. may have $W_{Li} > H_{Li}$. In yet other embodiments, a cut lens may be achieved by cutting the large lens elements of lens **204** to a height of e.g. 6.5 mm, 5 mm or 4 mm (in the Y direction), i.e. lens elements L_i that have height $H_{Li} > 6.5$ mm, 5 mm or 4.5 mm may be cut to 6.5 mm, 5 mm or 4.5 mm respectively.

Attention is now drawn to FIG. 2B, which depicts schematically another optical lens system disclosed herein and numbered **220**. Lens system **220** comprises a lens **204'** with a plurality of lens elements, optical element **206** (optional) and image sensor **208**. Ray tracing is provided as in FIG. 2A. Detailed optical data and surface data are given in Tables 5,

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6, 7 and 8. Table 8 provides details on the maximum CRAs of lens system **220**. Notation and field are identical to Table 4.

FIG. 2B shows 3 fields with 3 rays for each: the upper marginal-ray, the lower marginal-ray and the chief-ray.

The focusing range of lens system **220** is from infinity to 100 mm, i.e. the MIOD is 100 mm.

For focusing with lens **204'**, all lens elements are moved together. For changing focus from infinity to 100 mm, a lens stroke of 2.237 mm is required. For moving the lens, an actuator as known in the art may be used, e.g. a VCM. At the MIOD, lens system **210** achieves a magnification of -0.15 , corresponding to an object-image ratio of ca. 6.7:1. The HFOV decreases from 13.29 degrees when focused to infinity to 12.51 degrees when focused to the MIOD. Any focus position between infinity to 100 mm (as well as any magnification between 0 and -0.15) can be realized. For any object within the focus range, lens system **220** has a maximum field curvature (MFC) $<50\text{ }\mu\text{m}$.

Lens **204'** can be divided into two groups, a first focusing group which includes lens elements L_1 , L_2 , L_3 , L_4 and L_5 and a second CRA correction group which includes L_6 and L_7 . The focusing group and the CRA correction group are

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separated spatially from each other by a big gap of 4.960 mm. All lens elements of G1 together have an EFL1=14.08 mm, all lens elements of G2 together have an EFL2=13.94 mm. The TTL of lens **204'** is 18 mm.

FIG. 2H an embodiment numbered **260** of a lens system with a D-cut lens **204'-C** based on lens **204'**. As in the description of FIG. 2G, light loss is given as percentage of rays-through at coordinates (X, Y)=(0, 2.1 mm):

Infinity: 92.33%

1 m: 92.07%

0.5 m: 91.75%

20 cm: 90.76%

10 cm: 89.11%

D-cut lens **204'-C** is obtained by cutting the large lens elements of lens **204'** to a height of e.g. 6 mm (in the Y direction), i.e. the lens elements L_i of lens **204'** that have height $H_{Li}>6\text{ mm}$ (i.e. L_1 , L_6 and L_7) are cut to 6 mm. In other embodiments, a cut lens may be achieved by cutting the large lens elements of lens **204'** to a height of e.g. 6.5 mm, 5 mm or 4 mm (in the Y direction), i.e. the lens elements L_i that have height $H_{Li}>6.5\text{ mm}$, 5 mm or 4.5 mm may be cut to 6.5 mm, 5 mm or 4.5 mm respectively. For details on cut lenses it is referred to the description of FIG. 2A.

TABLE 5

Lens system 220									
EFL = 15.0 mm, F number = 2.2, Half FOV = 13.3 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S	Plano	Infinity	-1.030	3.408				
2	Lens 1	ASP	4.603	1.619	3.633	Plastic	1.54	55.91	119.97
3			4.335	1.382	3.441				
4	Lens 2	ASP	3.577	1.700	2.930	Plastic	1.54	55.91	9.38
5			9.836	0.571	2.729				
6	Lens 3	ASP	-25.617	0.260	2.710	Plastic	1.67	19.44	-5.17
7			4.059	0.651	2.619				
8	Lens 4	ASP	-5.649	0.671	2.605	Plastic	1.54	55.91	17.54
9			-3.704	0.040	2.518				
10	Lens 5	ASP	3.303	1.675	2.649	Plastic	1.67	19.44	13.22
11			4.172	4.960	2.471				
12	Lens 6	ASP	-3.581	1.519	3.048	Plastic	1.67	19.44	-35.15
13			-4.939	0.040	3.530				
14	Lens 7	ASP	12.341	1.703	3.857	Plastic	1.54	55.91	11.86
15			-12.986	0.780	3.890				
16	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.17	
17			Infinity	0.220	—				
18	Image	Plano	Infinity	—	—				

TABLE 6

Aspheric Coefficients					
Surface #	Conic	A4	A6	A8	A10
2	0	-2.62E-03	9.87E-06	-3.09E-07	-4.05E-07
3	0	-9.81E-03	4.21E-04	-2.87E-05	4.48E-07
4	0	-7.71E-03	-4.01E-05	-4.65E-05	1.34E-06
5	0	9.24E-03	-1.27E-03	-1.66E-05	2.67E-06
6	0	-1.68E-03	7.18E-05	1.10E-04	-1.03E-05
7	0	-2.43E-02	2.85E-03	-8.38E-05	-7.26E-06
8	0	1.45E-02	6.75E-04	-3.20E-05	-4.50E-06
9	0	1.54E-02	1.39E-03	-1.53E-04	1.25E-05
10	0	-6.69E-03	-1.41E-04	2.93E-05	-4.92E-06
11	0	-6.80E-03	1.38E-03	-1.76E-04	5.48E-06
12	0	7.57E-03	1.80E-04	4.85E-06	-5.12E-07
13	0	2.74E-03	-1.10E-05	2.79E-05	-1.63E-06

TABLE 6-continued

Aspheric Coefficients					
Surface #	Conic	A4	A6	A8	A10
14	0	-1.70E-03	-5.47E-05	1.09E-05	-2.45E-07
15	0	-2.92E-04	-1.37E-04	5.95E-06	1.00E-07

TABLE 7

Lens system 220 Variation of lens properties with object distance			
Object Distance [mm]	BFL [mm]	HFOV [deg]	Magnification
Infinity	1.210	13.29	0.00
1000	1.432	13.22	-0.02
500	1.655	13.14	-0.03
300	1.952	13.06	-0.05
250	2.101	13.01	-0.06
200	2.324	12.93	-0.08
150	2.698	12.80	-0.10
100	3.447	12.51	-0.15

TABLE 8

CRA/FOV-Lens system 220			
Object [mm]	Max CRA	HFOV (Image Height ("IH") = 3.5 mm) [deg]	Max CRA/FOV (IH = 3.5 mm)
Infinity	3.90	13.29	0.15
1000	4.54	13.22	0.17
500	5.07	13.14	0.19
300	4.65	13.06	0.18
250	4.74	13.01	0.18
200	4.87	12.93	0.19
150	5.04	12.80	0.20
100	5.28	12.51	0.21

FIG. 2C depicts schematically another optical lens system disclosed herein and numbered **230**. Lens system **230** comprises a lens **204"**, optical element **206** (optional) and image sensor **208**. Ray tracing is provided as in FIG. 2A-2B. Detailed optical data and surface data are given in Tables 9, 10, 11 and 12. Table 12 provides details on the maximum CRAs of lens system **220**. Notation and field are identical to Table 4.

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FIG. 2C shows 3 fields with 3 rays for each: the upper marginal-ray, the lower marginal-ray and the chief-ray. The focusing range of lens system **230** is from infinity to 76.4 mm, i.e. the MIOD is 76.4 mm. For focusing with lens **204"**, all lens elements are moved together. For changing focus from Infinity to 100 mm, a lens stroke of 2.881 mm is required. For moving the lens, an actuator (e.g. a VCM) as known in the art may be used. At the MIOD, lens system **230** achieves a magnification of -0.195, corresponding to an object-image ratio of ca. 5.1:1. The HFOV decreases from 13.29 degrees when focused to infinity to 12.52 degrees when focused to the MIOD. Any focus position between Infinity to 76.4 mm (as well as any magnification between 0 and -0.195) can be realized. For any object within the focus range, lens system **230** has a MFC<50 μ m.

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Lens **204"** can be divided into two groups, a first focusing group that includes L_1, L_2, L_3, L_4 and L_5 and a second CRA correction group that includes L_6, L_7 and L_8 . The focusing group and the CRA correction group are separated spatially from each other by a big gap of 3.839 mm. All lens elements of G1 together have an EFL1=13.50 mm, all lens elements of G2 together have an EFL2=11.85 mm. The TTL of lens **204"** is 18 mm.

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FIG. 2I shows an embodiment numbered **270** of a lens system with a cut lens **204"-C** based on lens **204"**. As in the description of FIG. 2G, light loss is given as percentage of rays-through at coordinates (X, Y)=(0, 2.1 mm):

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Infinity: 93.21%

1 m: 92.99%

0.5 m: 92.74%

20 cm: 92.01%

30

7.64 cm: 85.82%

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204"-C is obtained by cutting the large lens elements of lens **204"** to a height of e.g. 6 mm (in Y direction). The lens elements L_i of lens **204"** that have height $H_{Li}>6$ mm (i.e. L_1, L_2, L_6, L_7 and L_8) are cut to 6 mm. In other embodiments, a cut lens may be achieved by cutting the large lens elements of lens **204"** to a height of e.g. 6.5 mm, 5 mm or 4 mm (in the Y direction). For details on cut lenses it is referred to the description of FIG. 2A.

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TABLE 9

Lens system 230 EFL = 15.0 mm, F number = 2.2, HFOV = 13.29 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S	Plano	Infinity	-1.160	3.409				
2	Lens 1	ASP	5.187	1.335	3.629	Plastic	1.54	56.02	138.46
3			5.064	0.950	3.445				
4	Lens 2	ASP	3.563	1.513	3.150	Plastic	1.54	55.91	8.07
5			15.708	0.609	2.903				
6	Lens 3	ASP	-39.685	0.617	2.873	Plastic	1.65	21.78	-4.50
7			3.201	0.303	2.583				
8	Lens 4	ASP	-11.948	0.675	2.580	Plastic	1.53	55.97	22.38
9			-6.098	0.040	2.440				

TABLE 9-continued

Lens system 230 EFL = 15.0 mm, F number = 2.2, HFOV = 13.29 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
10	Lens 5	ASP	3.209	1.680	2.444	Plastic	1.67	19.44	10.03
11			4.813	3.839	2.212				
12	Lens 6	ASP	-3.377	1.002	2.840	Plastic	1.65	21.78	-6.59
13			-17.249	0.040	3.403				
14	Lens 7	ASP	-15.688	1.652	3.496	Plastic	1.58	30.11	9.50
15			-4.267	0.040	3.690				
16	Lens 8	ASP	12.159	1.701	4.123	Plastic	1.54	56.02	13.86
17			-18.946	1.592	4.063				
18	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.17	
19			Infinity	0.202	—				
20	Image	Plano	Infinity	0.000	—				

TABLE 10

Aspheric Coefficients					
Surface #	Conic	A4	A6	A8	A10
2	0	-1.26E-03	2.06E-05	-1.02E-06	-3.50E-07
3	0	-5.16E-03	3.31E-04	-2.35E-05	2.89E-07
4	0	-4.34E-03	2.48E-04	-4.84E-05	1.77E-06
5	0	8.67E-03	-8.16E-04	-1.20E-05	4.10E-06
6	0	-3.47E-04	2.29E-05	5.60E-05	-4.03E-06
7	0	-3.01E-02	2.92E-03	-7.26E-05	-6.05E-06
8	0	1.89E-02	2.99E-04	-6.15E-05	-8.51E-07
9	0	2.09E-02	2.15E-03	-3.43E-04	9.74E-06
10	0	-1.05E-02	-5.90E-04	1.64E-04	-1.37E-05
11	0	-5.32E-03	7.82E-04	-2.81E-05	7.36E-06
12	0	9.47E-04	5.64E-04	6.09E-05	-2.44E-06
13	0	-3.85E-03	1.06E-04	4.63E-05	-2.32E-06
14	0	3.81E-04	-3.62E-05	1.72E-05	-7.92E-07
15	0	3.94E-03	4.70E-05	-9.66E-06	5.03E-07
16	0	5.83E-05	-8.90E-05	5.22E-06	4.24E-08
17	0	-1.59E-03	1.52E-04	-1.02E-05	4.46E-07

TABLE 11

Lens system 230 Variation of lens properties with object distance			
Object Distance [mm]	BFL [mm]	HFOV [deg]	Magnification
Infinity	2.004	13.29	0.000
1000	2.225	13.24	-0.015
500	2.447	13.11	-0.030
300	2.742	13.18	-0.050
250	2.889	13.12	-0.060
200	3.110	13.02	-0.075
150	3.477	12.92	-0.100
100	4.209	12.69	-0.149
76.4	4.885	12.52	-0.195

TABLE 12

CRA/FOV-Lens system 230			
Object [mm]	Max CRA	HFOV (Image height = 3.5 mm)	Max CRA/FOV (Image Height = 3.5 mm)
Infinity	2.57	13.29	0.10
1000	3.21	13.24	0.12
500	3.72	13.11	0.14

TABLE 12-continued

CRA/FOV-Lens system 230			
Object [mm]	Max CRA	HFOV (Image height = 3.5 mm)	Max CRA/FOV (Image Height = 3.5 mm)
300	3.35	13.18	0.13
250	3.45	13.12	0.13
200	3.59	13.02	0.14
150	3.78	12.92	0.15
100	4.06	12.69	0.16
76	3.99	12.52	0.16

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Attention is now drawn to FIGS. 2D-2F, which depict schematically yet another optical lens system disclosed herein and numbered **240**. Lens system **240** comprises a lens **204''**, optical element **206** (optional) and image sensor **208**. Ray tracing is provided as in FIG. 2A-2C. Detailed optical data and surface data are given in Tables 13, 14, 15, 16 and 17. Table 17 provides details on the Max CRAs of lens system **240**. Notation and field are identical to Table 4. FIGS. 2D-2F show three fields with 3 rays for each: the upper marginal-ray, the lower marginal-ray and the chief-ray.

FIG. 2D shows lens system **240** focused to infinity ("Config. A"), FIG. 2E shows lens system **240** focused to 100 mm ("Config. B") and FIG. 2F shows lens system **240**

focused to 50 mm ("Config. C"). The object-lens distance focused on is given by Surface 0 in Table 15. The focusing range of lens system **240** is from infinity to 50 mm, i.e. the MIOD is 50 mm.

Lens **204'''** is divided into two groups that move relative to each other for focusing. A first lens group ("G1") includes lens elements L_1 and L_2 , and a second lens group ("G2") includes L_3 , L_4 , L_5 and L_6 . The big gap of between G1 and G2 decreases from 2.625 mm when focused to Infinity to 1.946 mm when focused to 100 mm, and to 1.303 mm when focused to 50 mm (see Surface #5 in table 15). For focusing, lens **204'''** also moves as one unit so that the BFL changes (see Surface #13 in Table 15).

For moving the lens, an actuator (e.g. VCM) may be used, At the MIOD, lens system **240** achieves a magnification of -0.40 , corresponding to an object-image ratio of ca. 2.5:1. The HFOV decreases from 13.1 degrees when focused to infinity to 9.3 degrees when focused to the MIOD (see Table 16). Any focus position between Infinity to 50 mm (as well as any magnification between 0 and -0.4) can be realized.

For any object within the focus range, lens system **240** has a MFC $<50\text{ }\mu\text{m}$. All lens elements of G1 together have an EFL1=38.61 mm, all lens elements of G2 together have an EFL2=15.36 mm. The TTL of lens **204'''** is 15.8 mm.

FIG. 2J shows an embodiment numbered **280** of a lens system with a cut lens **204'''**-C based on lens **204'''**. As in the description of FIG. 2G, light loss is given as percentage of rays-through at coordinates (X, Y)=(0, 2.1 mm):

Infinity: 94.58%

10 cm: 91.53%

5 cm: 91.02%

Cut lens **204'''**-C is obtained by cutting the large lens elements of lens **204'''** to a height of 6 mm (in Y direction), i.e. the lens elements L_i of lens **204'''** that have height $H_{Li}>6$ mm (L_1 , L_2 , L_5 and L_6) are cut to 6 mm. In other embodiments, a cut lens may be achieved by cutting the large lens elements of lens **204'''** to a height of e.g. 6.5 mm, 5 mm or 4 mm (in the Y direction).

TABLE 13

Lens system 240									
EFL = 15.0 mm, F number = 2.2, HFOV = 13.1 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S	Plano	Infinity	-1.138	3.394				
2	Lens 1	ASP	5.548	1.713	3.399	Plastic	1.53	55.66	10.11
3			-225.776	0.031	3.268				
4	Lens 2	ASP	4.454	1.000	3.084	Plastic	1.64	23.52	-9.96
5			2.401	2.625	2.627				
6	Lens 3	ASP	4.197	1.000	2.722	Plastic	1.53	55.66	15.25
7			7.888	4.910	2.661				
8	Lens 4	ASP	-6.881	0.435	2.717	Plastic	1.66	20.37	1841.21
9			-7.016	0.162	2.848				
10	Lens 5	ASP	2.335	0.551	2.906	Plastic	1.53	55.66	-24.20
11			1.816	1.274	3.339				
12	Lens 6	ASP	-7.488	1.251	3.393	Plastic	1.53	55.66	20.05
13			-4.675	0.344	3.546				
14	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.17	
15			Infinity	0.290	—				
16	Image	Plano	Infinity	—	—				

For details on cut lenses it is referred to the description of FIG. 2A.

TABLE 14

Aspheric Coefficients					
Surface #	Conic	A4	A6	A8	A10
2	-4.21E-01	6.50E-04	-1.01E-05	1.56E-06	-5.69E-08
3	-2.24E+02	2.36E-03	-1.58E-04	5.38E-06	-7.57E-08
4	-4.52E+00	-5.19E-04	5.31E-05	-6.85E-06	3.53E-07
5	-2.24E+00	2.21E-03	1.61E-05	-1.54E-06	7.95E-07
6	-8.98E-03	-7.71E-05	-5.52E-05	5.85E-06	-8.17E-07
7	6.54E-01	6.97E-04	-2.34E-05	-6.13E-07	-5.58E-07
8	-6.59E+01	4.55E-03	-1.90E-03	2.62E-05	3.32E-06
9	-4.32E-01	1.47E-02	-3.66E-03	2.57E-04	-5.76E-06
10	-4.54E+00	-2.21E-02	-1.28E-03	3.91E-04	-2.22E-05
11	-1.73E+00	-3.64E-02	4.33E-03	-2.94E-04	6.72E-06
12	-2.40E+01	1.32E-03	3.07E-04	-3.67E-05	-4.21E-07
13	-1.13E+00	6.63E-04	-3.99E-04	6.48E-05	-3.13E-06

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TABLE 15

Lens system 240 Variation of surface thicknesses			
Surface #	Config. A	Config. B	Config. C
0	1.00E+06	100	50
5	2.625	1.946	1.303
13	0.344	2.941	6.318

TABLE 16

Lens system 240		
Config. #	HFOV	Magnification
A	13.1 deg	0
B	11.3 deg	-0.17
C	9.3 deg	-0.40

TABLE 17

CRA/FOV-Lens system 240			
Object [mm]	Max CRA	HFOV (Image Height ("IH") = 3.5 mm) [deg]	Max CRA/FOV (IH = 3.5 mm)
Infinity	12.65	13.09	0.48
1000	12.51	11.64	0.54
50	11.44	9.92	0.58

TABLE 18

EFL ratios between groups G1 and G2				
Embodiment	Last surface of first group	EFL1 (EFL of G1)	EFL2 (EFL of G2)	Ratio EFL1/EFL2
200	9	14.71	13.55	1.09
220	11	14.08	13.94	1.01
230	11	13.50	11.85	1.14

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TABLE 18-continued

EFL ratios between groups G1 and G2				
Embodiment	Last surface of first group	EFL1 (EFL of G1)	EFL2 (EFL of G2)	Ratio EFL1/EFL2
240	5	38.61	15.36	2.51
290	11	14.1	12.3	1.15

Table 18 shows an overview on the EFLs of all lens elements of the G1 and G2 respectively as well as ratios EFL1/EFL2 for lens system embodiments **200**, **220**, **230**, **240** and **290**.

FIG. 2K depicts schematically another optical lens system disclosed herein and numbered **290**. Lens system **290** comprises a lens **204'''** with a plurality of lens elements, optical element **206** (optional) and image sensor **208**. Ray tracing is provided as in FIG. 2A. Detailed optical data, surface data and further lens properties are given in Tables **19**, **20**, **21** and **22**.

The focusing range of lens system **290** is from infinity to 52 mm (MIOD=52 mm).

For focusing with lens **204'''**, all lens elements are moved together. For changing focus from infinity to 52 mm, a lens stroke of 4.507 mm is required. At the MIOD, lens system **290** achieves a magnification of -0.29, corresponding to an object-image ratio of ca. 3.4:1. The HFOV decreases from 9.57 degrees when focused to infinity to 8.52 degrees when focused to the MIOD. Any focus position between infinity to 52 mm (as well as any magnification between 0 and -0.29) can be realized. For any object within the focus range, lens system **290** has a MFC<50 μ m.

Lens **204'''** can be divided into two groups, a first focusing group that includes L_1 , L_2 , L_3 , L_4 and L_5 and a second CRA correction group that includes L_6 , L_7 and L_8 . The focusing group and the CRA correction group are separated spatially from each other by a big gap of 3.974 mm. All lens elements of G1 together have an EFL1=14.1 mm, all lens elements of G2 together have an EFL2=12.3 mm. The TTL of lens **204'''** is 18.6 mm.

Some embodiments may include a cut lens based on lens **204'''**. The cut lens may be achieved by cutting the large lens elements of lens **204'''** to a height of e.g. 6.5 mm, 5 mm or 4 mm (in the Y direction). For details on cut lenses, it is referred to the description of FIG. 2A.

TABLE 19

Embodiment 290 F = 15.5 mm, F number = 2.58, HFOV = 13.29 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S		Infinity	-0.732	3.000				
2	Lens 1	ASP	5.487	1.243	3.000	Plastic	1.54	56.02	178.01
3			5.351	0.942	3.000				
4	Lens 2	ASP	3.687	1.573	3.000	Plastic	1.54	55.91	8.24
5			17.247	0.659	3.000				
6	Lens 3	ASP	-34.749	0.716	2.969	Plastic	1.65	21.78	-4.57
7			3.302	0.310	2.644				
8	Lens 4	ASP	-13.318	0.684	2.641	Plastic	1.53	55.97	24.85
9			-6.772	0.041	2.494				
10	Lens 5	ASP	3.284	1.758	2.490	Plastic	1.67	19.44	9.94
11			5.040	3.974	2.228				
12	Lens 6	ASP	-3.602	1.073	2.897	Plastic	1.65	21.78	-7.06
13			-18.159	0.044	3.000				
14	Lens 7	ASP	-14.373	1.615	3.000	Plastic	1.58	30.11	10.02
15			-4.342	0.041	3.000				
16	Lens 8	ASP	14.157	1.757	3.000	Plastic	1.54	56.02	14.47
17			-17.049	1.743	3.000				

TABLE 19-continued

Embodiment 290									
F = 15.5 mm, F number = 2.58, HFOV = 13.29 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
18	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.17	
19			Infinity	0.212	—				
20	Image	Plano	Infinity	—	—				

TABLE 20

Aspheric Coefficients						
Surface #	Conic	A4	A6	A8	A10	A12
2	0	-1.21E-03	-6.93E-05	3.32E-05	-6.30E-06	5.62E-07
3	0	-5.07E-03	3.57E-04	-1.87E-05	-2.46E-06	4.29E-07
4	0	-4.06E-03	5.18E-04	-1.20E-04	1.22E-05	-8.59E-07
5	0	7.39E-03	-8.92E-05	-2.83E-04	6.07E-05	-6.11E-06
6	0	-4.46E-04	3.79E-04	-1.33E-04	3.83E-05	-5.13E-06
7	0	-2.57E-02	1.10E-06	1.63E-03	-5.53E-04	9.45E-05
8	0	1.88E-02	-2.83E-03	1.86E-03	-5.87E-04	9.51E-05
9	0	1.91E-02	2.54E-03	-8.30E-04	2.32E-04	-5.19E-05
10	0	-9.94E-03	4.29E-04	-4.14E-04	1.81E-04	-3.75E-05
11	0	-4.66E-03	1.18E-03	-3.00E-04	1.12E-04	-2.25E-05
12	0	7.88E-04	-4.64E-04	4.63E-04	-8.04E-05	7.21E-06
13	0	-2.84E-03	-9.77E-04	3.76E-04	-4.70E-05	2.74E-06
14	0	1.30E-03	-9.09E-04	2.99E-04	-4.17E-05	2.82E-06
15	0	4.93E-03	-8.53E-04	2.09E-04	-2.46E-05	1.42E-06
16	0	1.54E-03	-1.01E-03	1.89E-04	-1.87E-05	1.06E-06
17	0	-6.52E-04	-1.55E-04	1.55E-05	5.19E-07	-1.41E-07

Surface #	Conic	A14	A16
2	0	-2.71E-08	5.34E-10
3	0	-2.83E-08	6.99E-10
4	0	4.33E-08	-1.18E-09
5	0	3.31E-07	-7.68E-09
6	0	3.20E-07	-7.80E-09
7	0	-8.39E-06	3.03E-07
8	0	-7.83E-06	2.58E-07
9	0	6.16E-06	-2.88E-07
10	0	3.69E-06	-1.41E-07
11	0	2.26E-06	-6.86E-08
12	0	-3.08E-07	4.71E-09
13	0	-6.12E-08	-6.65E-11
14	0	-8.10E-08	4.86E-10
15	0	-3.30E-08	9.41E-11
16	0	-3.20E-08	3.98E-10
17	0	7.77E-09	-1.35E-10

TABLE 21

Embodiment 290			
Variation of lens properties with object distance			
Object Distance [mm]	BFL [mm]	HFOV [deg]	Magnification
Infinity	2.166	9.57	0.00
1033	2.395	9.56	-0.01
517	2.625	9.59	-0.03
310	2.931	9.60	-0.05
258	3.083	9.46	-0.06
207	3.312	9.42	-0.07
155	3.693	9.34	-0.10
103	4.453	9.15	-0.15
52	6.673	8.52	-0.29

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TABLE 22

CRA/FOV-Embodiment 290			
Object [mm]	Max. CRA [deg]	HFOV (Image Height ("IH") = 2.8 mm) [deg]	Max. CRA/FOV (IH = 2.8 mm)
Infinity	0.65	9.57	0.03
1033	0.96	9.56	0.05
517	0.47	9.59	0.02
310	0.47	9.60	0.02
258	1.82	9.46	0.10
207	2.05	9.42	0.11
155	2.37	9.34	0.13
103	2.86	9.15	0.16
52	3.74	8.52	0.22

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According to some examples, at least part of the lens elements can have a shape (profile) in cross-section (in plane X-Y, which is orthogonal to the optical lens system and which generally coincides with the optical axis) that is not circular. In particular, as shown for example in FIG. 7 for a lens barrel 700 carrying lens elements 702, at least some of the lens elements can have a width W_{Li} (measured along axis X) which is greater than their height H_{Li} (measured along axis Y). Barrel 700 further comprises a barrel wall (or surrounding) 704. The height H_{Li} can correspond to the total height of the lens element (including the mechanical part). In some embodiments, a lens element in lens barrel 700 may have a symmetry about axis Y and/or about axis X. A non-circular lens barrel such as 700 may be shaped according to the cut lens elements of a lens such as e.g. lens 204-C. As shown in FIG. 7, the height of a lens barrel may be only slightly higher than the lens element having the largest height in the lens. For example, a lens barrel may be 0.05 mm higher than the highest lens element. A lens barrel having an identical height as the highest lens element is described for example in co-owned international patent application PCT/IB2018/050988, which is incorporated herein by reference in its entirety.

According to some examples, W_{Li} is substantially greater than H_{Li} (for example, by at least a percentage which is equal or greater than 20%, these values being not limiting). In some examples, W_{Li} may be greater than H_{Li} by a percentage of 20-70%. Consider lens element La of folded lens 204' as an example: W_{L8} is greater than H_{L8} by a percentage of 32%.

It is appreciated that certain features of the presently disclosed subject matter, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the presently disclosed subject matter, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination.

Furthermore, for the sake of clarity the term “substantially” is used herein to imply the possibility of variations in values within an acceptable range. According to one example, the term “substantially” used herein should be interpreted to imply possible variation of up to 10% over or under any specified value. According to another example, the term “substantially” used herein should be interpreted to imply possible variation of up to 5% over or under any specified value. According to a further example, the term “substantially” used herein should be interpreted to imply possible variation of up to 2.5% over or under any specified value.

Unless otherwise stated, the use of the expression “and/or” between the last two members of a list of options for selection indicates that a selection of one or more of the listed options is appropriate and may be made.

It should be understood that where the claims or specification refer to “a” or “an” element, such reference is not to be construed as there being only one of that element.

All references mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual reference was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present disclosure.

What is claimed is:

1. A folded digital camera, comprising:

a lens system with a lens and an image sensor, the lens having $N \geq 6$ lens elements L_i , wherein a first lens element L_1 faces an object, and an effective focal length (EFL); and

an optical path folding element (OPFE) for providing a folded optical path between the object and the lens, wherein the lens system has a focusing range that covers object-lens distances from infinity to a minimal object distance (MIOD),

wherein $MIOD/EFL < 20$,

wherein the lens is divided into two lens groups,

wherein the lens elements within each lens group do not move with respect to each other,

wherein the two lens groups move with respect to each other for focusing to any object-lens distance within the focusing range,

wherein the lens has a total track length (TTL) and wherein $TTL/EFL < 1.5$, and

wherein for the focusing to any object-lens distance within the focusing range, each of the two lens groups also move with respect to the image sensor.

2. The folded digital camera of claim 1, wherein the two lens groups are separated by a gap having a gap thickness, and wherein for changing focus from infinity to an object-lens distance that fulfills $500 \text{ mm} \geq \text{object-lens distance} \geq 100 \text{ mm}$, the gap thickness changes by more than 0.5 mm and by less than 1 mm.

3. The folded digital camera of claim 2, wherein $10 \text{ mm} < EFL < 20 \text{ mm}$.

4. The folded digital camera of claim 1, wherein the two lens groups are separated by a gap having a gap thickness, and wherein for changing focus from infinity to an object-lens distance that fulfills $500 \text{ mm} \geq \text{object-lens distance} \geq 50 \text{ mm}$, the gap thickness changes by more than 1 mm and by less than 2 mm.

5. The folded digital camera of claim 4, wherein $10 \text{ mm} < EFL < 20 \text{ mm}$.

6. The folded digital camera of claim 1, wherein $MIOD/EFL < 15$.

7. The folded digital camera of claim 1, wherein $MIOD/EFL < 10$.

8. The folded digital camera of claim 1, wherein $MIOD < 500 \text{ mm}$ and wherein $10 \text{ mm} < EFL < 30 \text{ mm}$.

9. The folded digital camera of claim 1, wherein $MIOD < 300 \text{ mm}$ and wherein $10 \text{ mm} < EFL < 30 \text{ mm}$.

10. The folded digital camera of claim 1, wherein $MIOD < 250 \text{ mm}$ and wherein $10 \text{ mm} < EFL < 30 \text{ mm}$.

11. The folded digital camera of claim 1, wherein $MIOD < 150 \text{ mm}$ and wherein $12.5 \text{ mm} < EFL < 20 \text{ mm}$.

12. The folded digital camera of claim 1, wherein the camera has an f number < 4 .

13. The folded digital camera of claim 1, wherein the camera has an f number < 3 .

14. The folded digital camera of claim 1, wherein the camera has an f number < 2.5 .

15. The folded digital camera of claim 1, wherein the camera has an f number < 2.25 .

16. The folded digital camera of claim 1, further comprising a voice coil motor based focusing mechanism for focusing the camera.

17. The folded digital camera of claim 1, further comprising an optical element, wherein the optical element is located between the lens and the image sensor.

18. The folded digital camera of claim 1, wherein the folded digital camera is included in a mobile device.

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19. The folded digital camera of claim **18**, wherein the mobile device is a smartphone.

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